Filas Inteligentes: Trabajo Práctico Nro. 3

72.25 - Simulación de Sistemas

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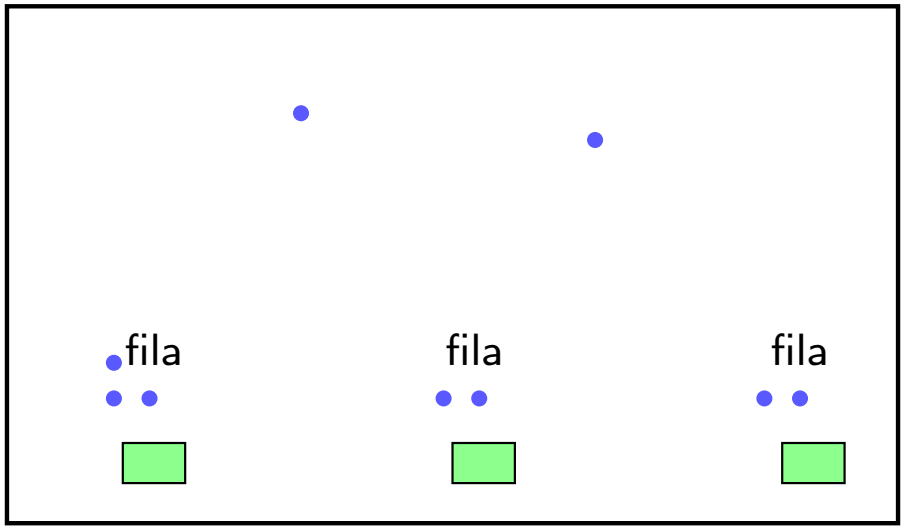
Introducción

Sistema Real

Los sistemas con múltiples servidores y filas aparecen en:

- Cajas de supermercado y centros de atención.
- Check-in, embarque y controles de acceso.
- Peajes y otros sistemas con congestión espacial.

Interesa comparar cómo la frecuencia de llegada, el tiempo de servicio y la geometría de la fila modifican la congestión y la permanencia de los clientes.



Modelo Matemático

Proceso de arribo:

$$\Delta t_{\text{arr}} \sim \text{Exp}\left(\frac{1}{t_1}\right)$$

Proceso de servicio:

$$\Delta t_{\text{serv}} \sim \text{Exp}\left(\frac{1}{t_2}\right)$$

Asignación Probabilística

Para cada servidor i se definió un costo temporal estimado:

$$c_i = \frac{d_i}{v} + n_i\, t_2$$

donde d_i es la distancia al servidor y n_i la carga asociada a su fila.
La asignación final se realiza con un sorteo ponderado:

$$P_i = \frac{e^{-c_i}}{\sum_j e^{-c_j}}$$

Modalidades:

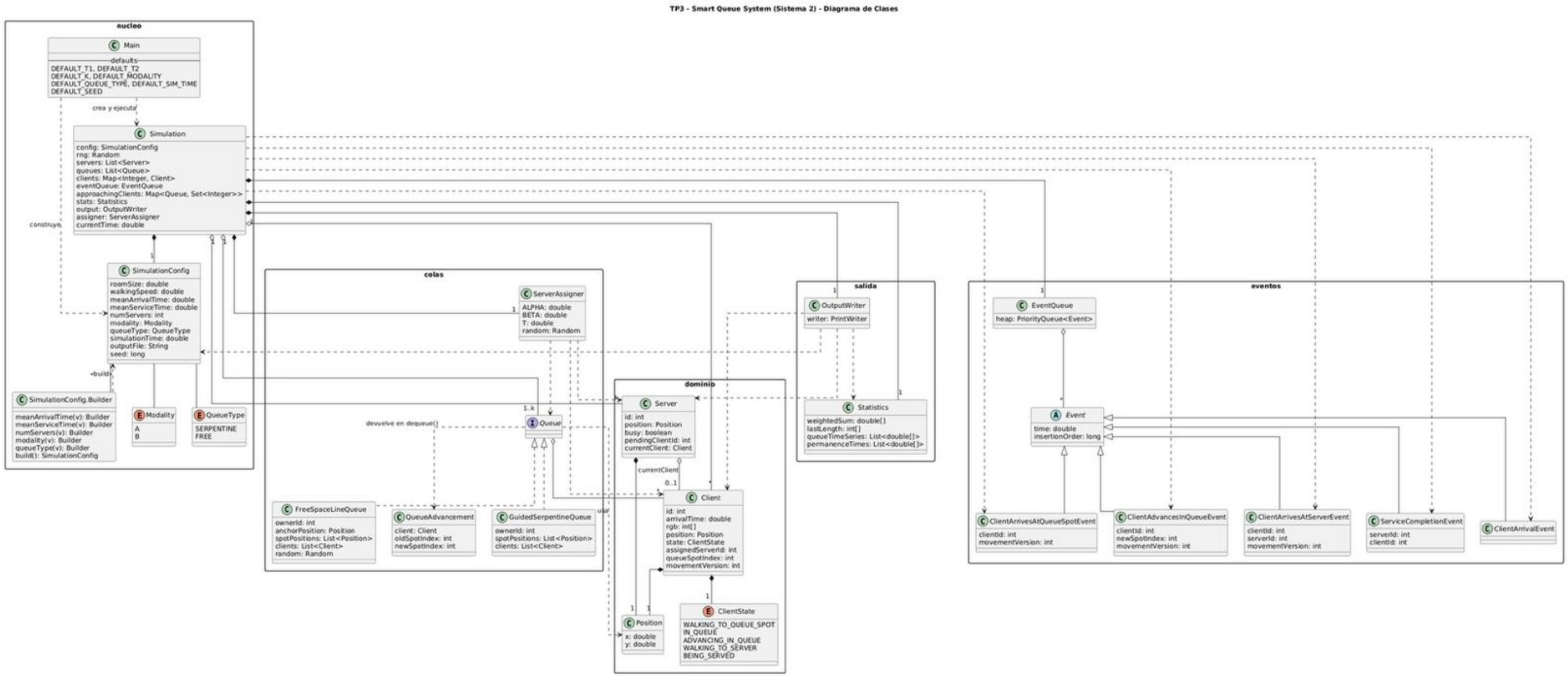
- A: una fila por servidor.
- B: una fila ũnica compartida.

Filas:

- Guiada: serpentina.
- Libre: heurística espacial.

Implementación

Arquitectura del Motor



El motor inicializa servidores y colas, agenda eventos y escribe el estado del sistema en cada evento.

Algoritmo Implementado

Algorithm Loop principal de simulación dirigida por eventos

```
Inicializar servidores, colas y estructura de eventos.
Agendar el primer arribo de cliente.
while haya eventos con t ≤ t_f do
    Tomar el próximo evento de la cola prioritaria.
    Actualizar posiciones interpoladas de clientes en movimiento.
    Procesar el evento correspondiente.
    Registrar el largo de fila en ese instante.
    Escribir el snapshot del sistema en el output.
end while
Calcular observables acumulados y escribir estadísticas finales.
```

Eventos concretos: arribo de cliente, llegada al lugar en la fila, avance en fila, llegada al servidor y fin de servicio.

Simulaciones

Parámetros Estudiados

Parámetros fijos del sistema:

- Recinto: $30 \times 30\,\mathrm{m}^2$.
- Velocidad de caminata: $1\,\mathrm{m/s}$.
- Tipo de fila: guiada, serpentina.
- $T_f = 1000\,\mathrm{s}$.

Parametros variables:

- $t_2 \in [1, 8]\ \mathrm{s}$
- $t_1 \in [0.3, 3]\ \mathrm{s}$
- $k \in [1, 8]$

Observables Definidos

Largo total de fila:

$$L(t) = \sum_{q=1}^{N_{\text{colas}}} \ell_q(t)$$

Promedio temporal en régimen estacionario:

$$\bar{L} = \frac{1}{t_f - t_{est}} \int_{t_{est}}^{t_f} L(t)\, dt$$

Tasa de crecimiento:

- $m =$ pendiente del ajuste lineal de la segunda mitad de $L(t)$. Si $m < 0.02$ clientes/s, se considera estable.

Observables Definidos

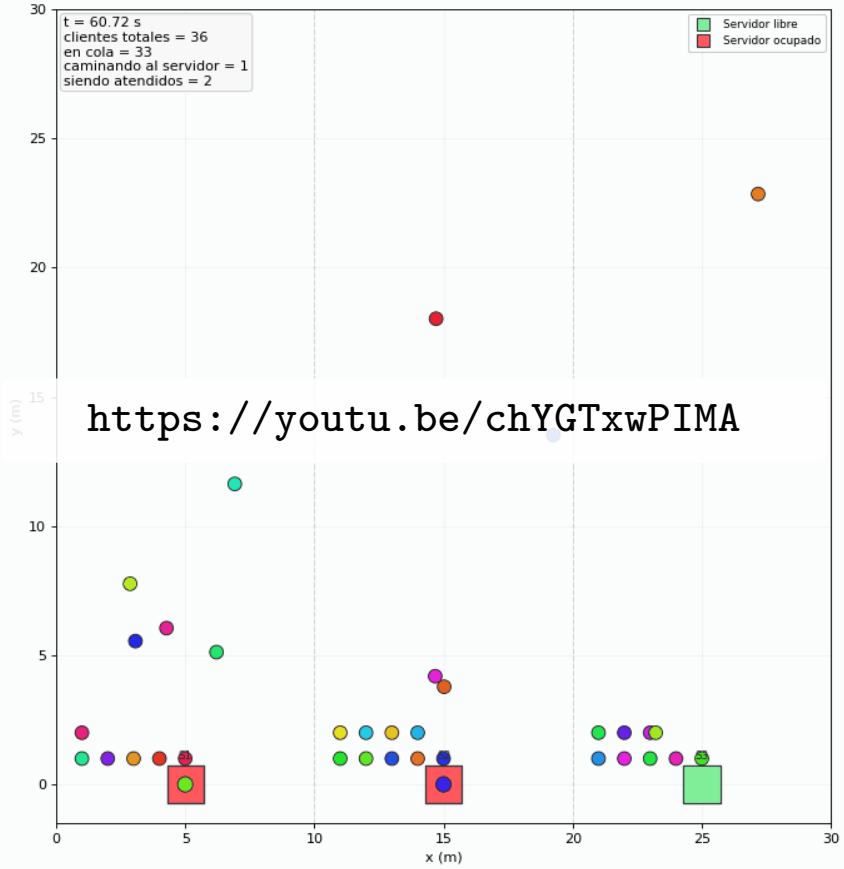
Distribución de permanencias:

$$\tau_i = t_i^{\text{salida}} - t_i^{\text{llegada}}$$

$$\bar{\tau} = \frac{1}{N_s} \sum_{i=1}^{N_s} \tau_i$$

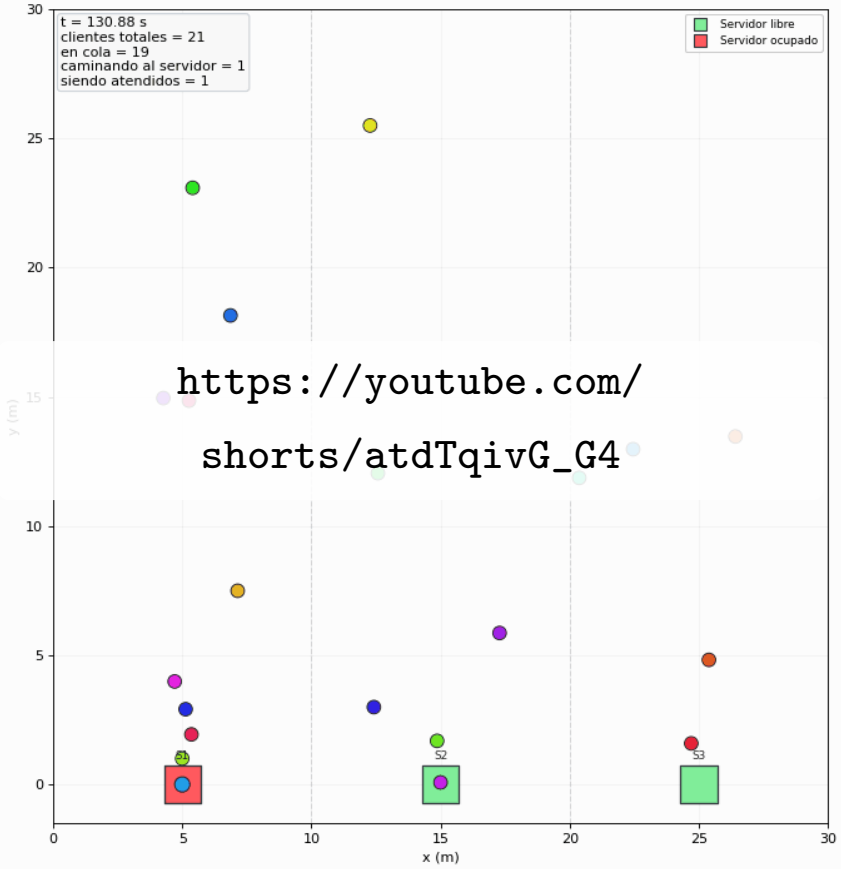
Resultados

Animaciones — Modalidad A



https://youtu.be/chYGTxwPIMA

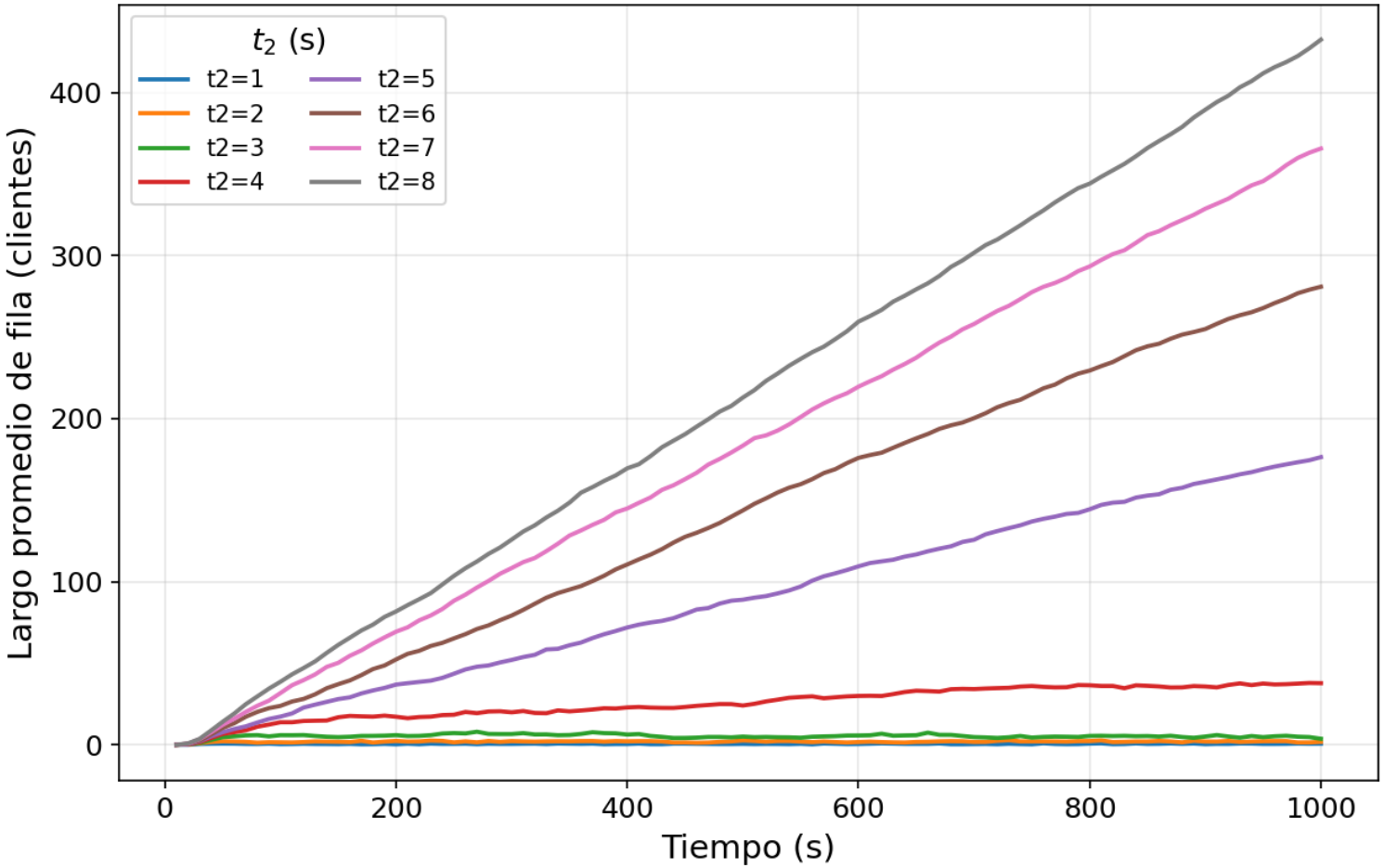
$t_1 = 1 \quad | \quad t_2 = 3.5 \quad | \quad k = 3$



https://youtube.com/shorts/atdTqivG_G4

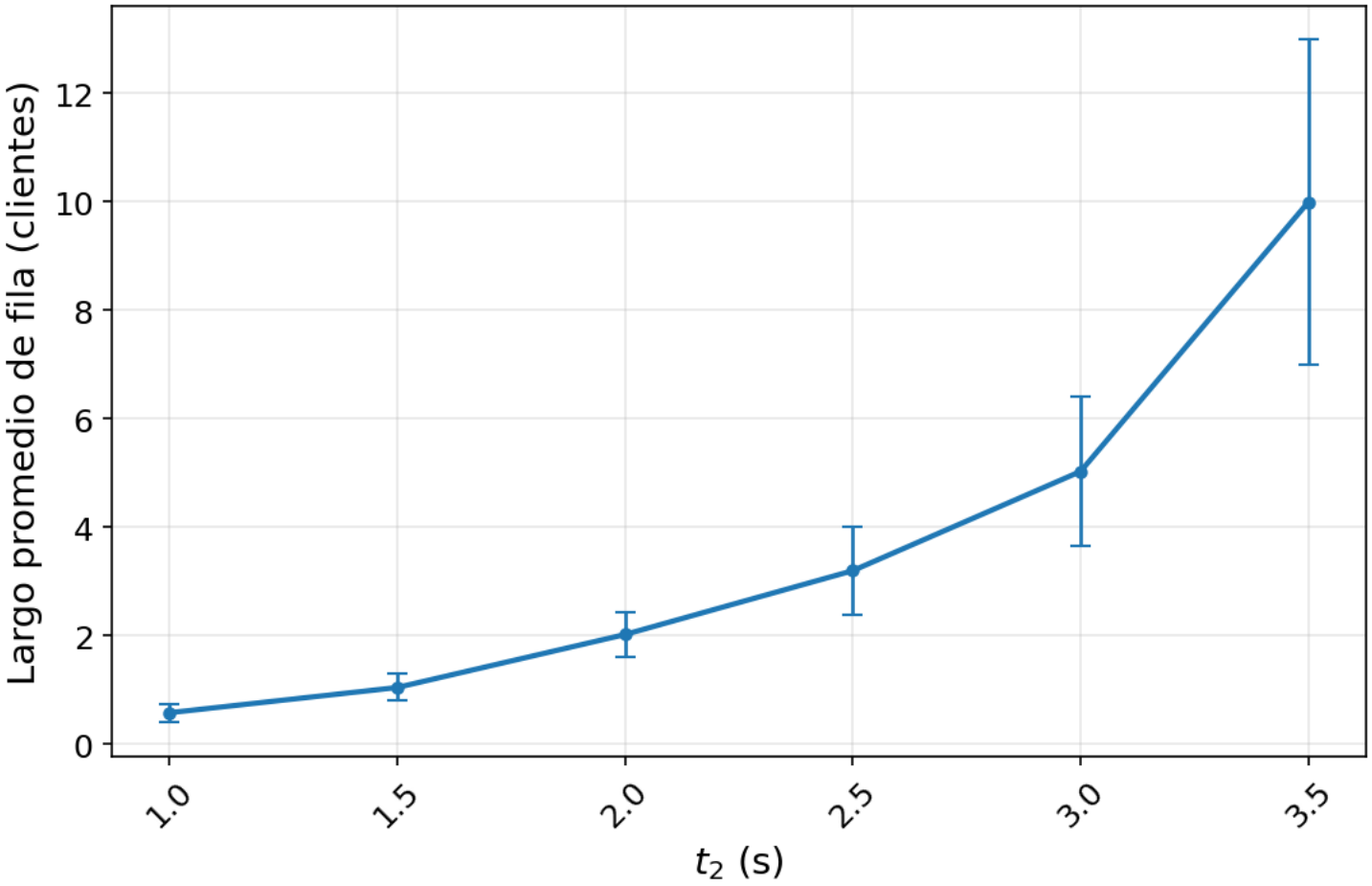
$t_1 = 1 \quad | \quad t_2 = 2 \quad | \quad k = 3$

Variación de t_2 — Modalidad A — $\langle L \rangle$ vs tiempo



$t_1 = 1\,\mathrm{s}$ | $k = 5$ | runs = 30

Variación de t_2 — Modalidad A — Largo de la fila



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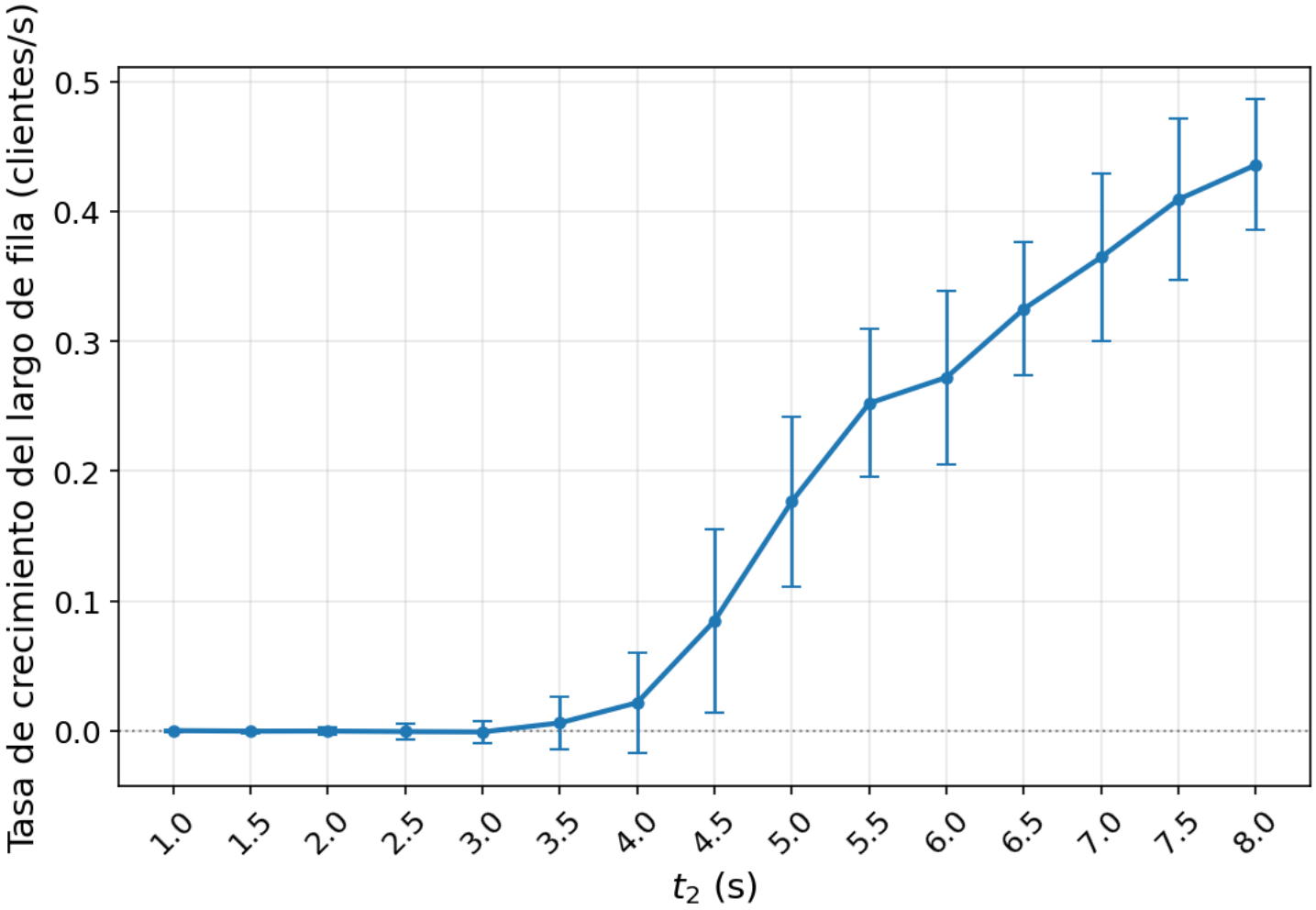
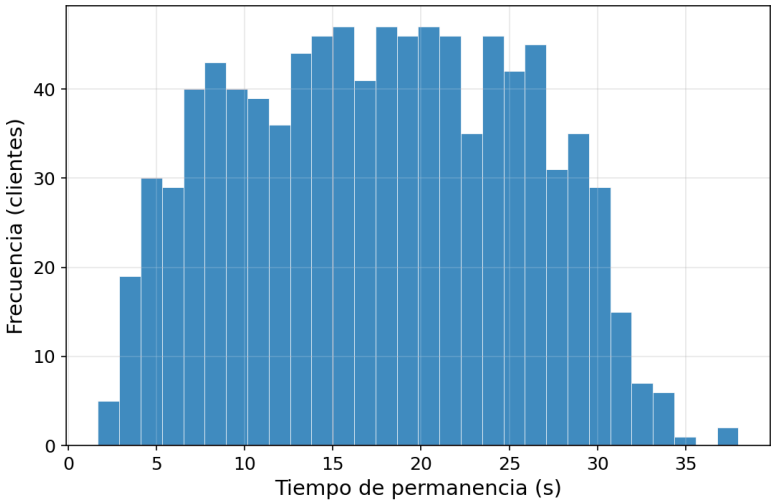
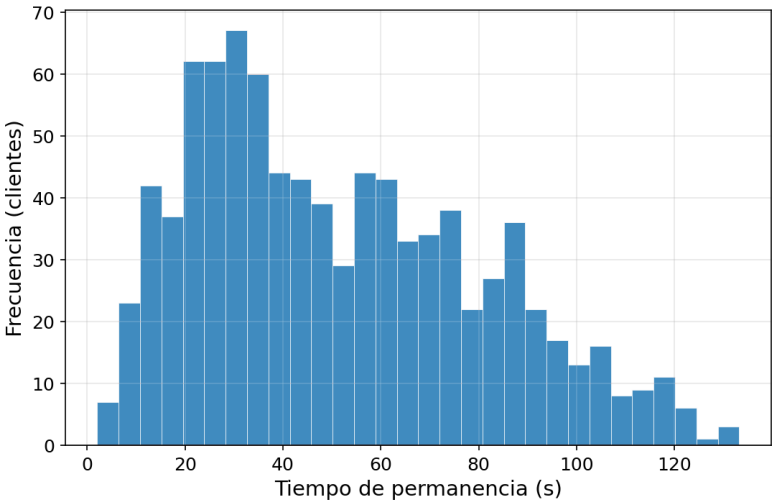
Variación de t_2 — Modalidad A — Tasa de crecimiento

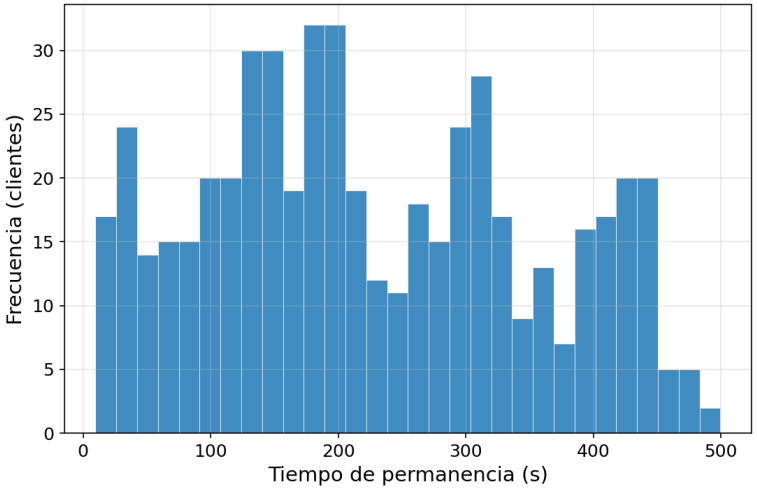


$t_1 = 1\,\mathrm{s}$ | $k = 5$ | runs = 30

Variación de t_2 — Modalidad A — Histogramas de permanencia

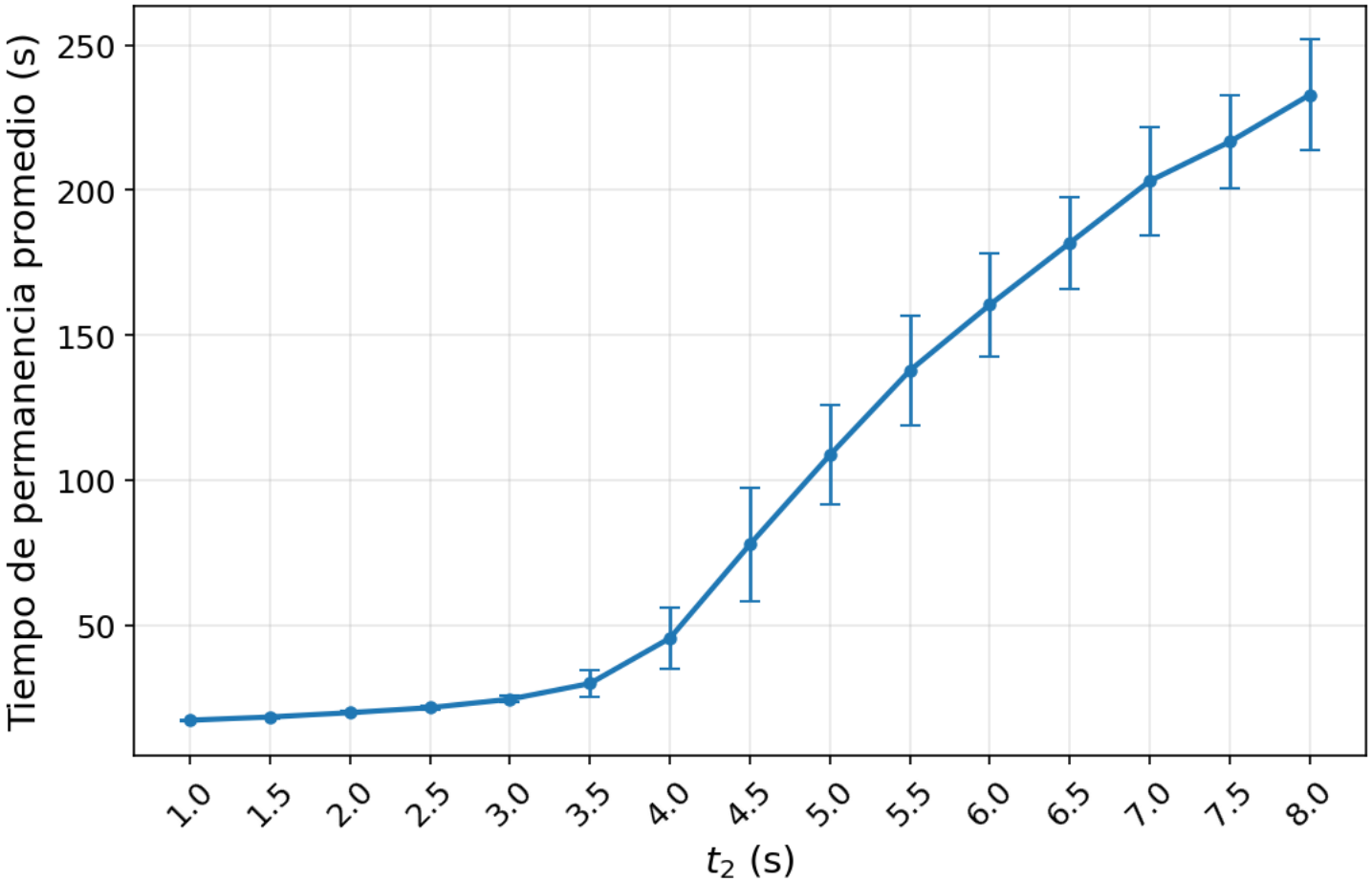






$t_1 = 1\,\mathrm{s} \mid k = 5 \mid t_2 \in \{1,\ 4,\ 8\}\,\mathrm{s}$
(izq. a der.)

Variación de t_2 — Modalidad A — Tiempo de permanencia



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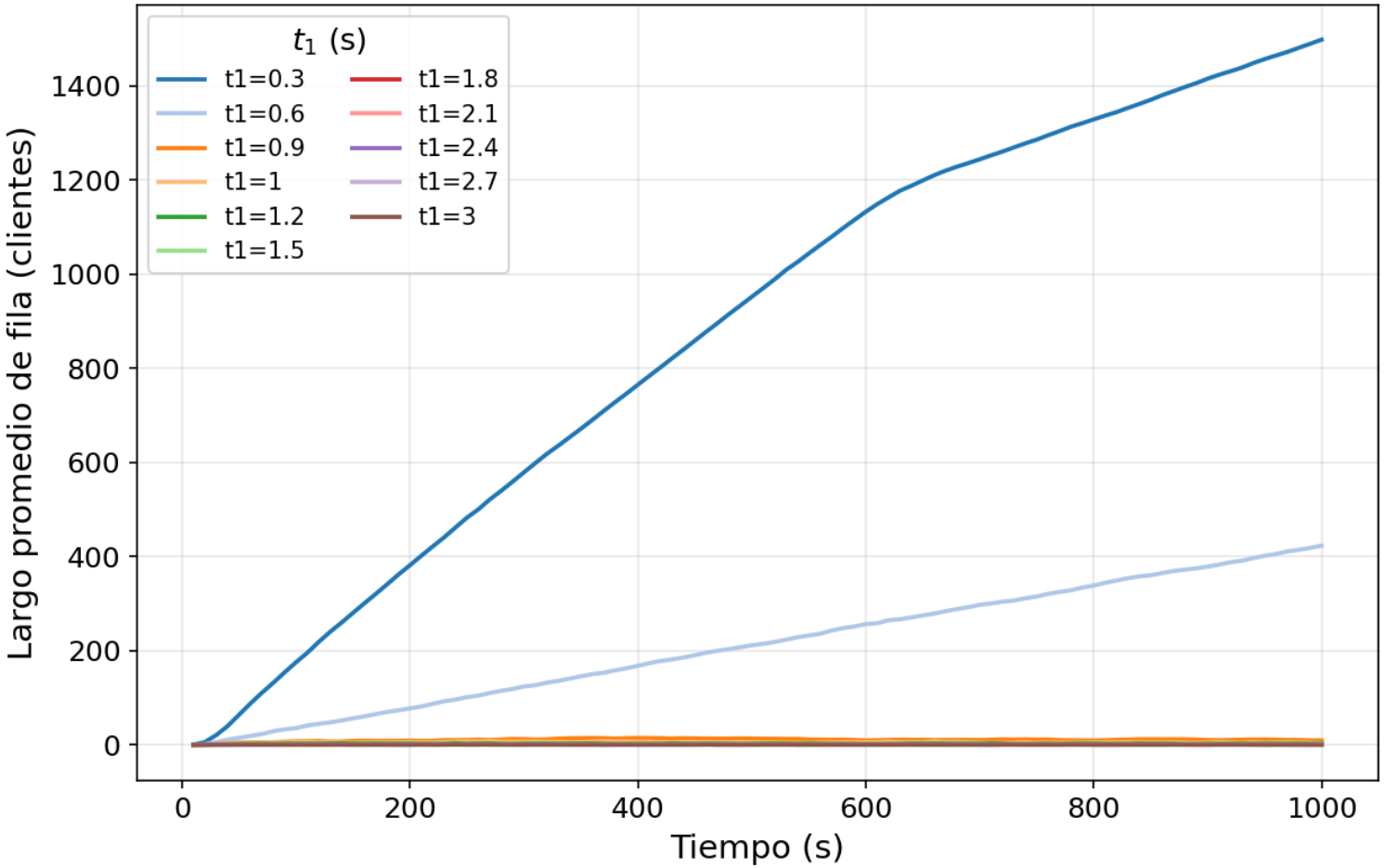
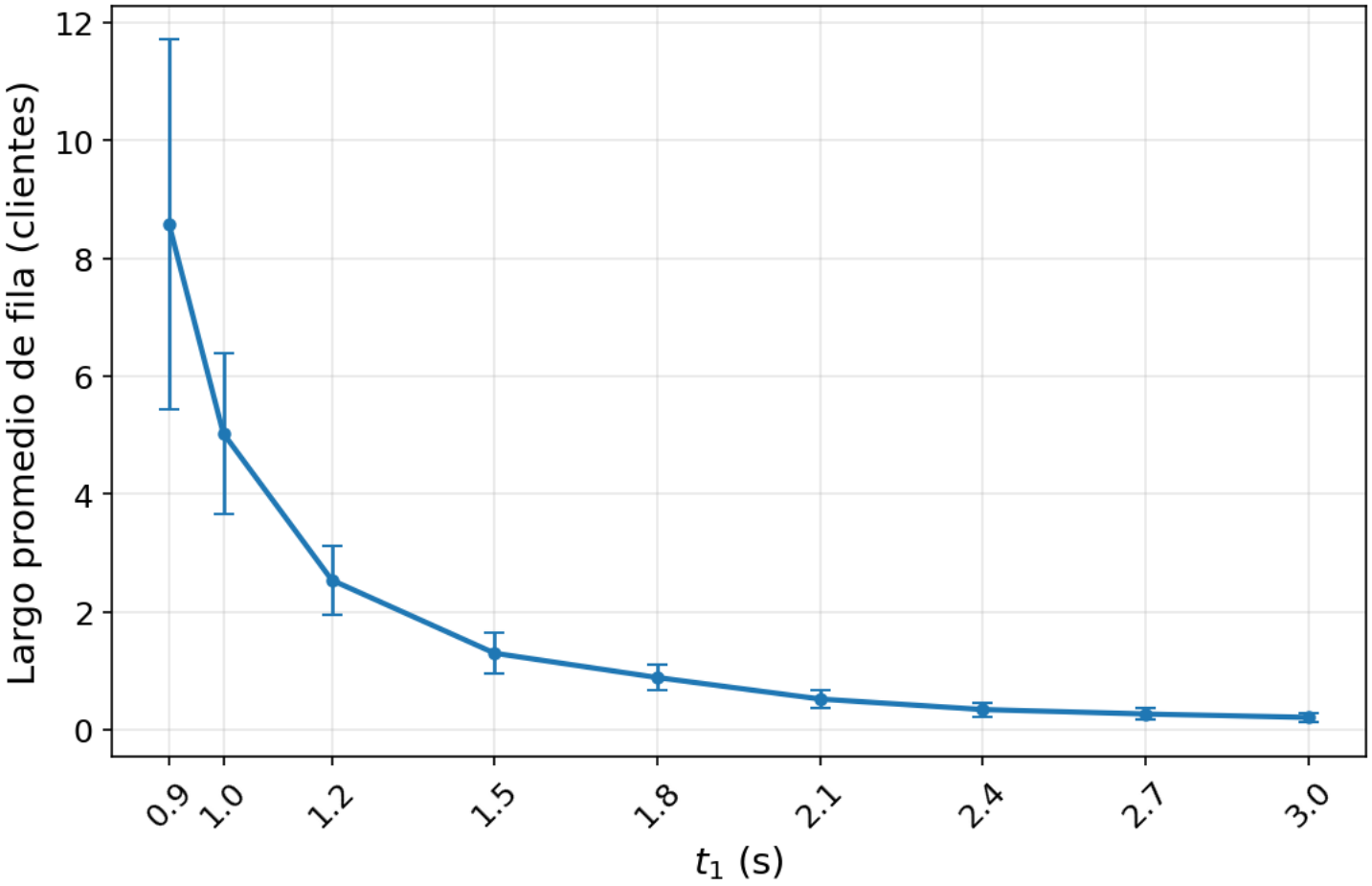
Variación de t_1 — Modalidad A — $\langle L \rangle$ vs tiempo



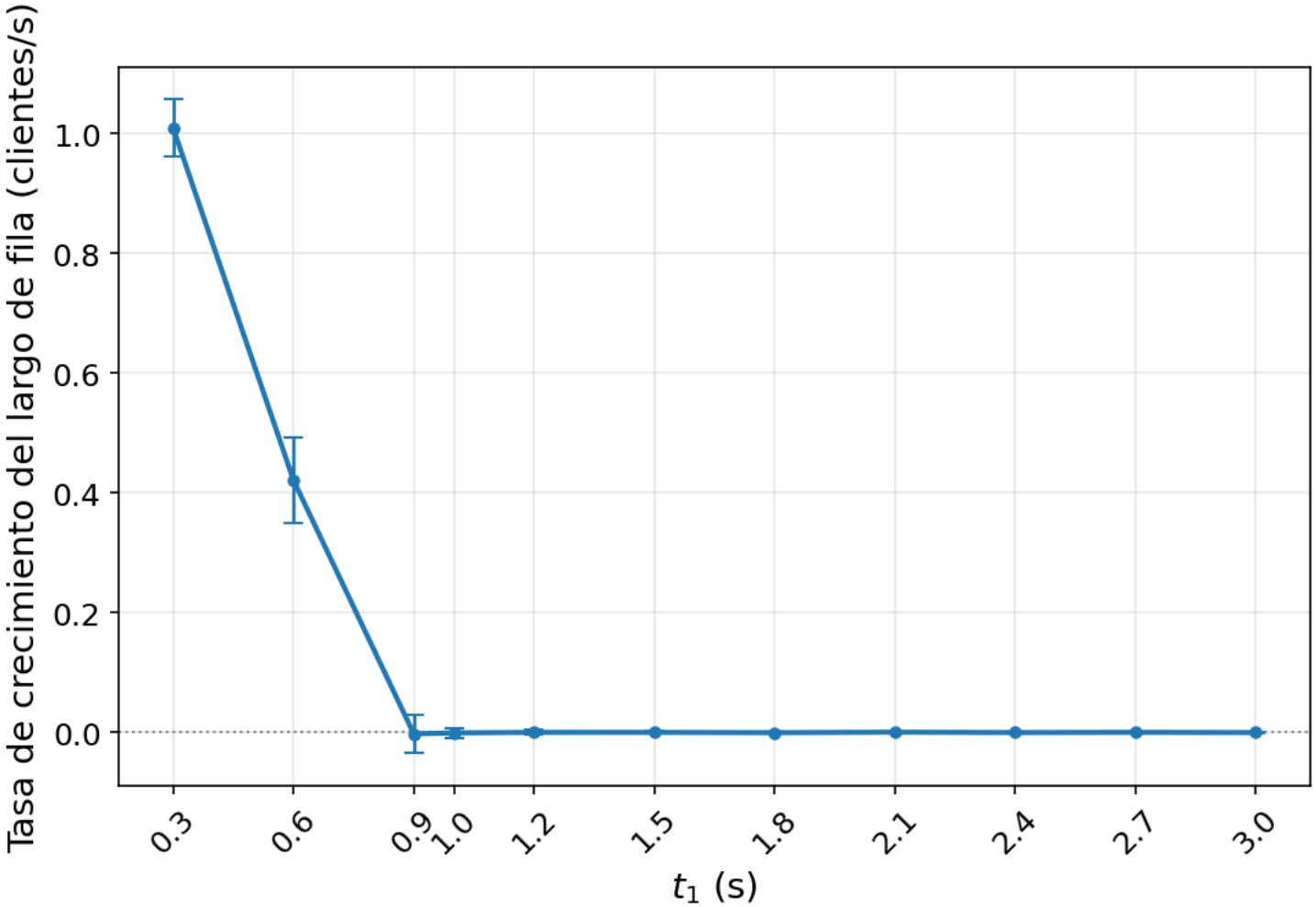
$t_2 = 3\,\mathrm{s}$ | $k = 5$ | runs = 30

Variación de t_1 — Modalidad A — Largo de la fila



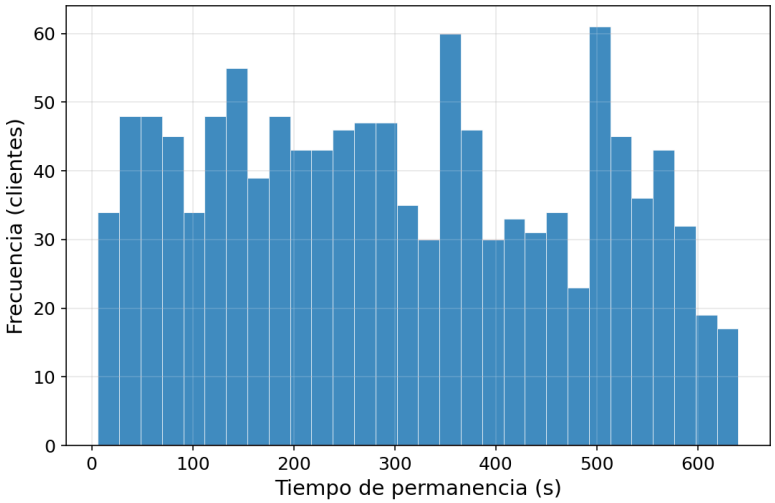
$t_2 = 3\,\mathrm{s}$ | $k = 5$ | runs = 30

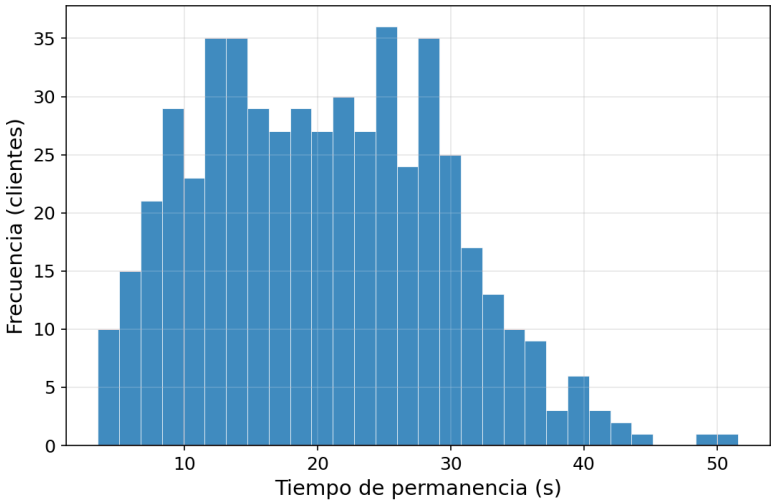
Variación de t_1 — Modalidad A — Tasa de crecimiento

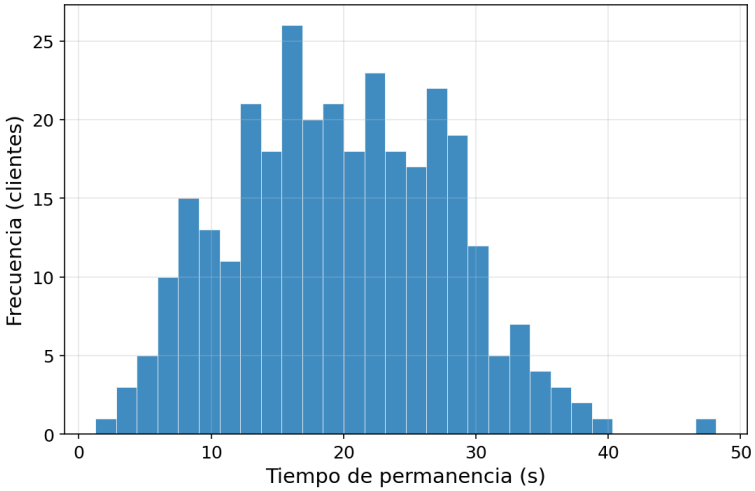


$t_2 = 3\,\mathrm{s}$ | $k = 5$ | runs = 30

Variación de t_1 — Modalidad A — Histogramas de permanencia

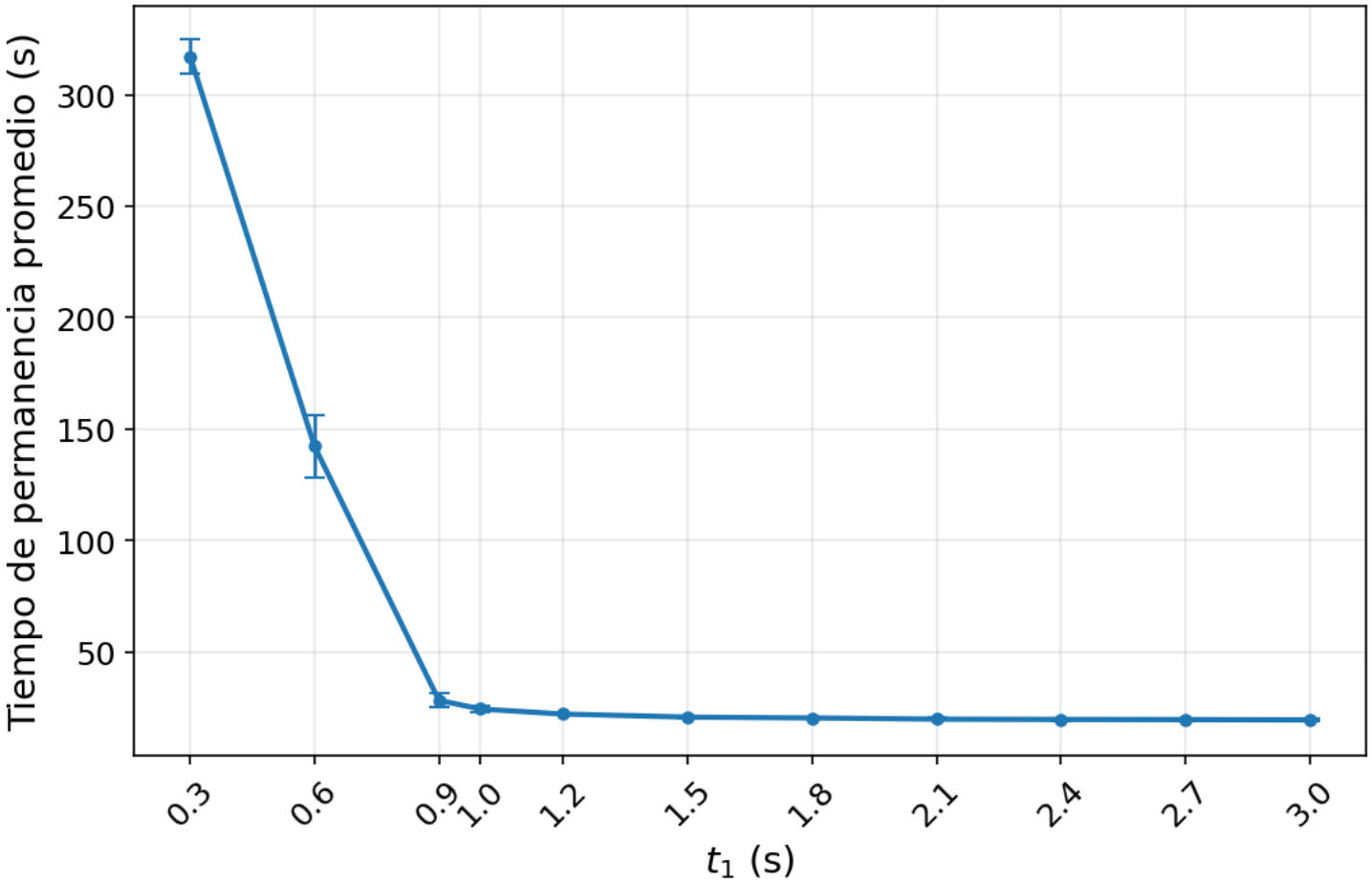






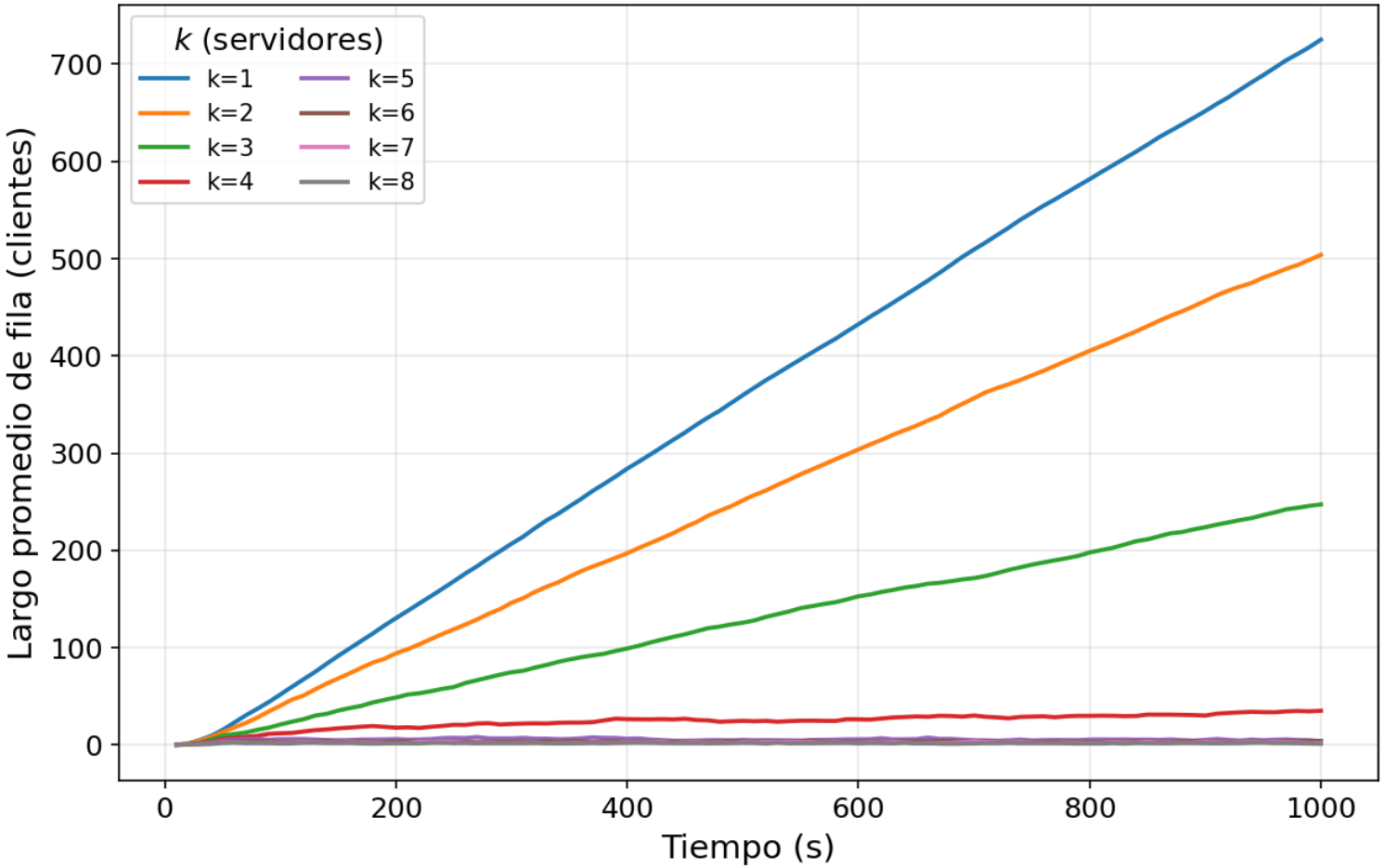
$t_2 = 3\,\mathrm{s}$ | $k = 5$ |
$t_1 \in \{0.3,\ 1.8,\ 3\}\,\mathrm{s}$ (izq. a der.)

Variación de t_1 — Modalidad A — Tiempo de permanencia



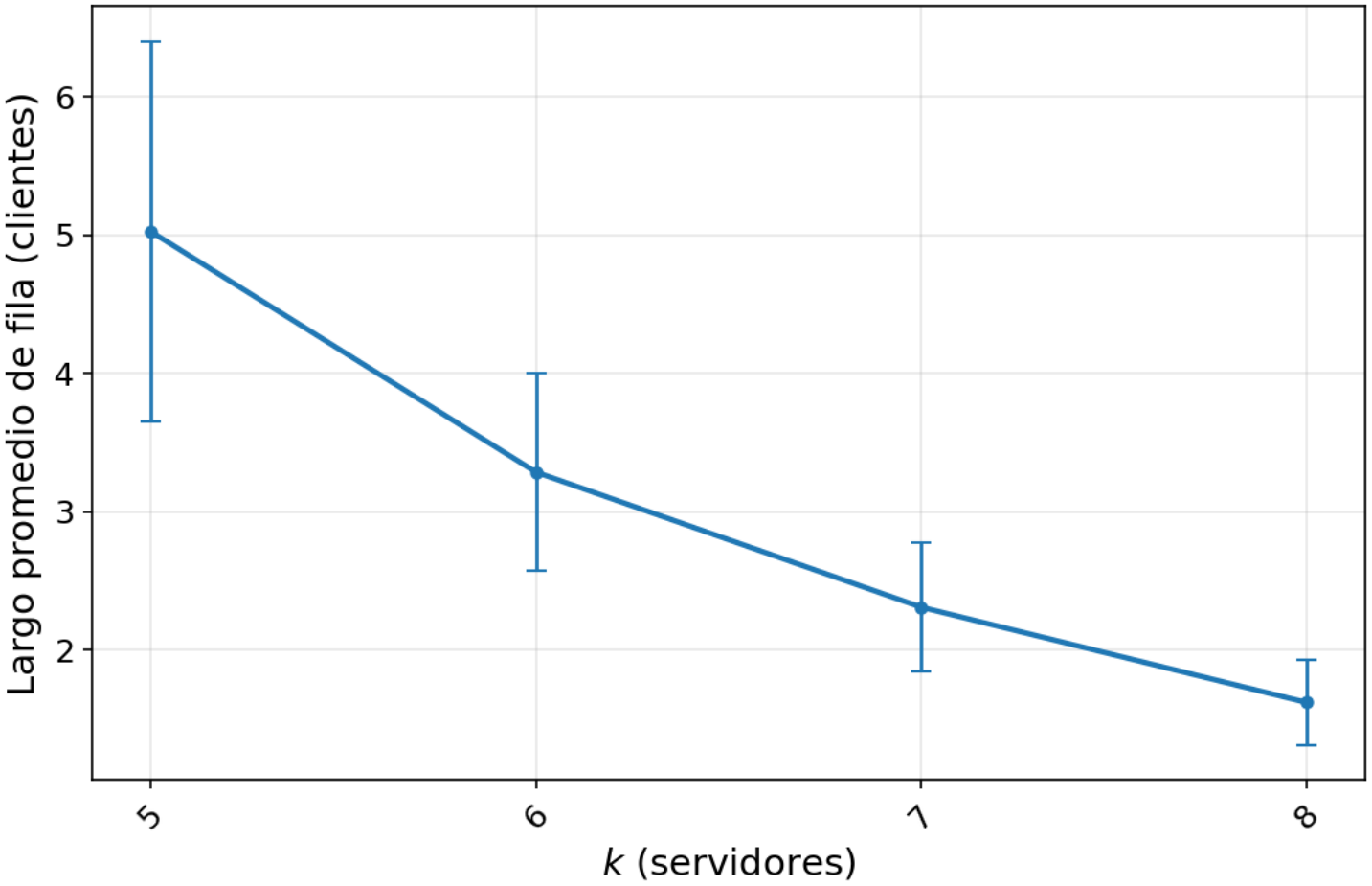
$t_2 = 3\,\mathrm{s}$ | $k = 5$ | runs = 30

Variación de k — Modalidad A — $\langle L \rangle$ vs tiempo



$t_1 = 1\,\mathrm{s}$ | $t_2 = 3\,\mathrm{s}$ | runs = 30

Variación de k — Modalidad A — Largo de la fila



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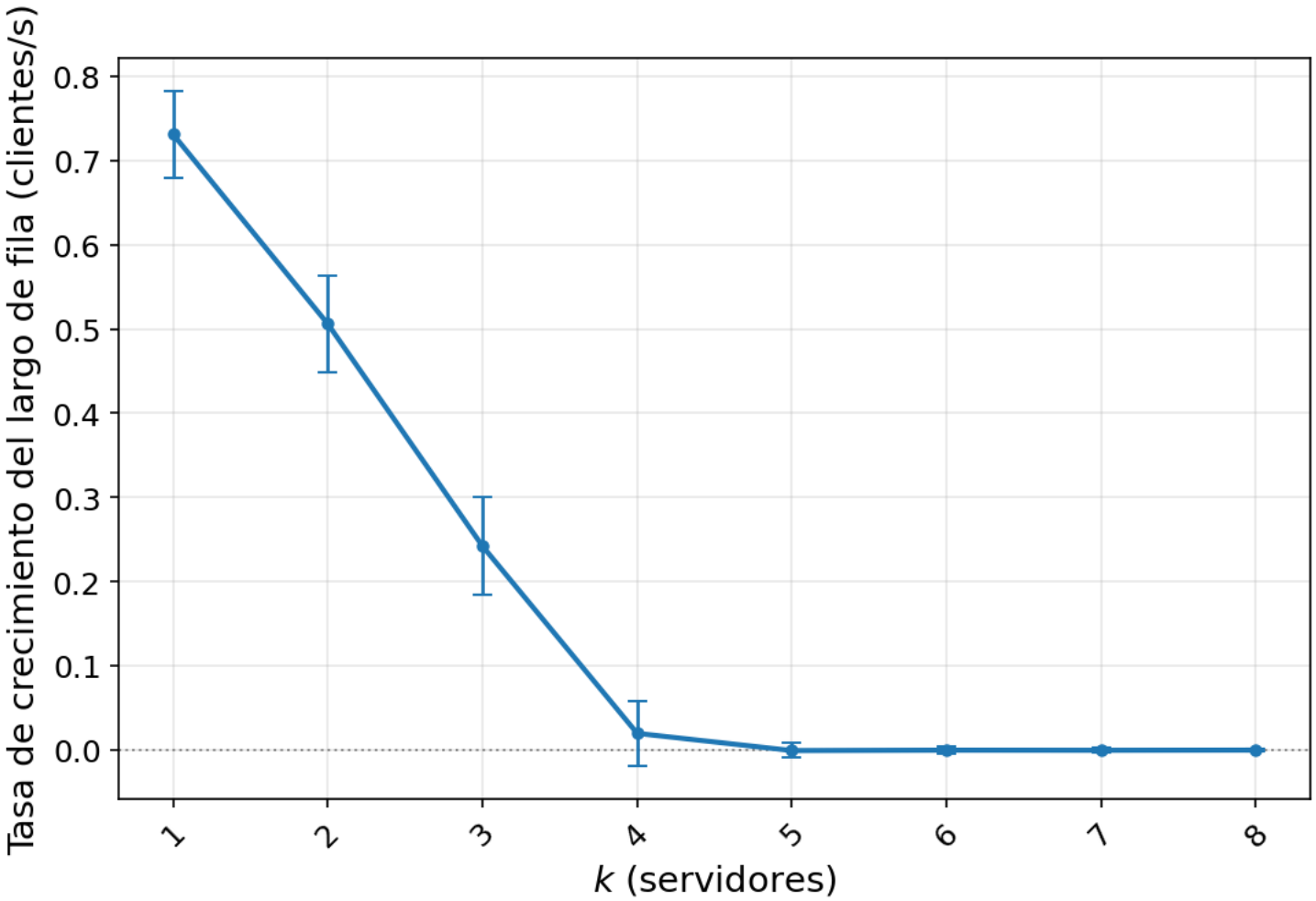
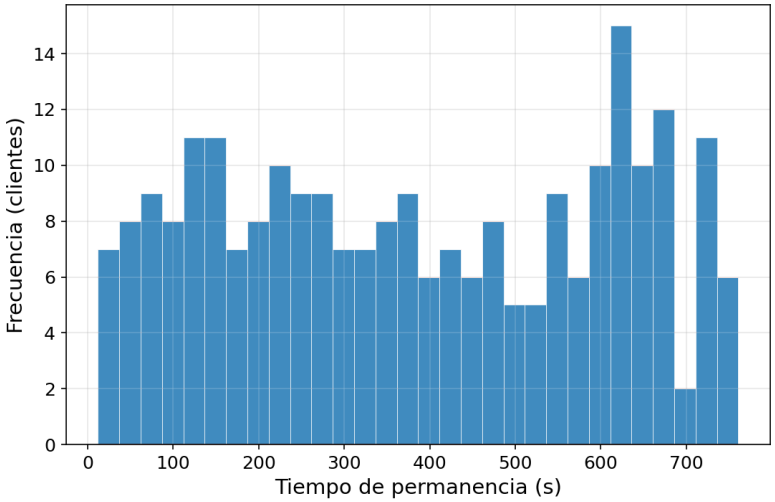
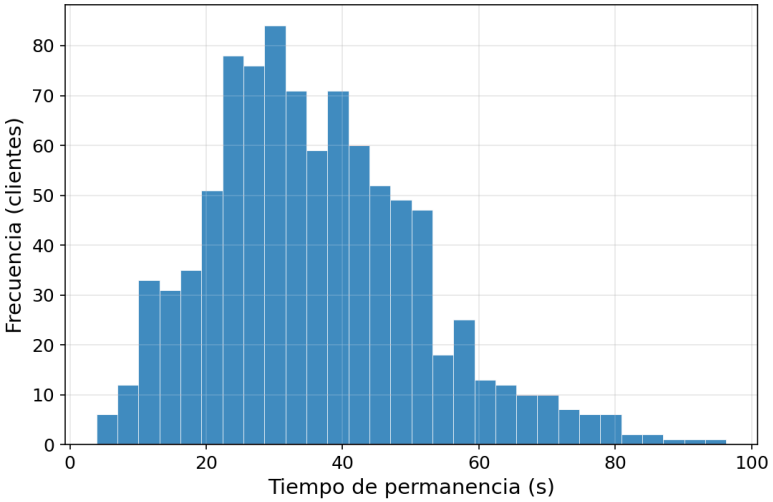
Variación de k — Modalidad A — Tasa de crecimiento

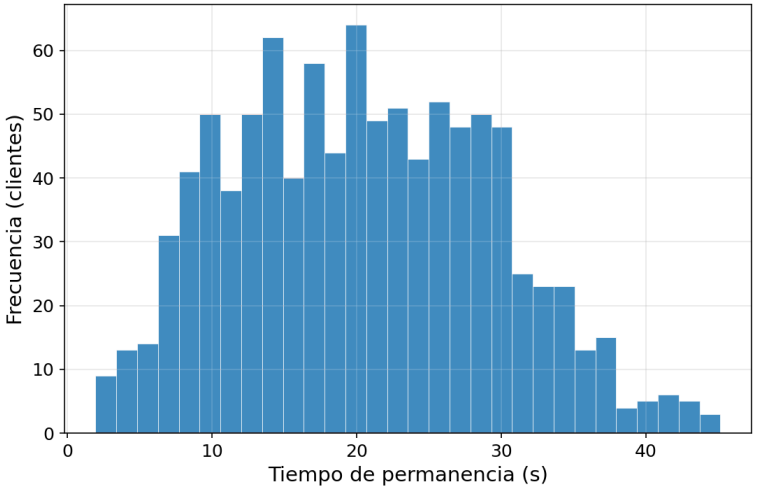


$t_1 = 1\,\mathrm{s}$ | $t_2 = 3\,\mathrm{s}$ | runs = 30

Variación de k — Modalidad A — Histogramas de permanencia

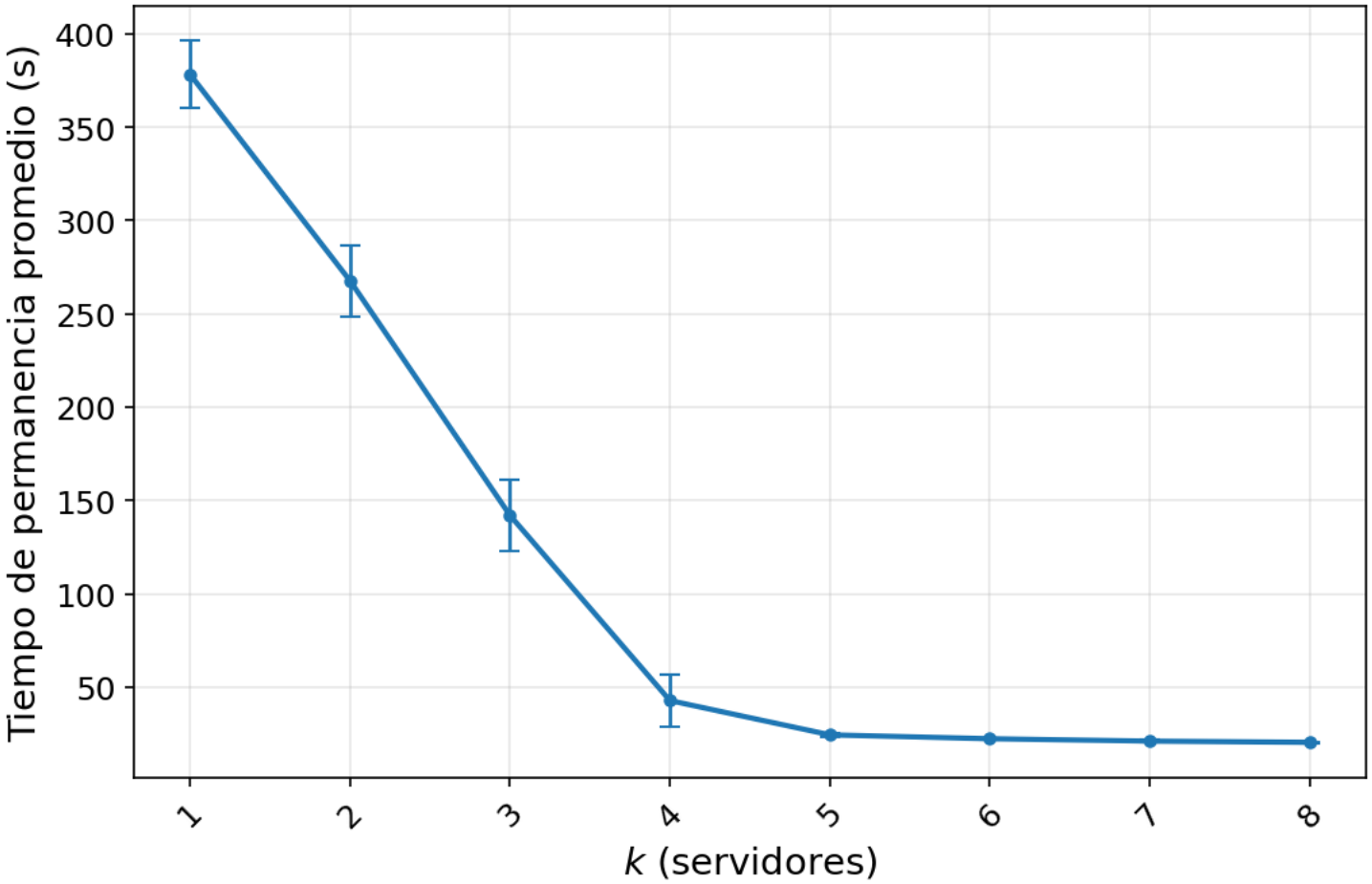






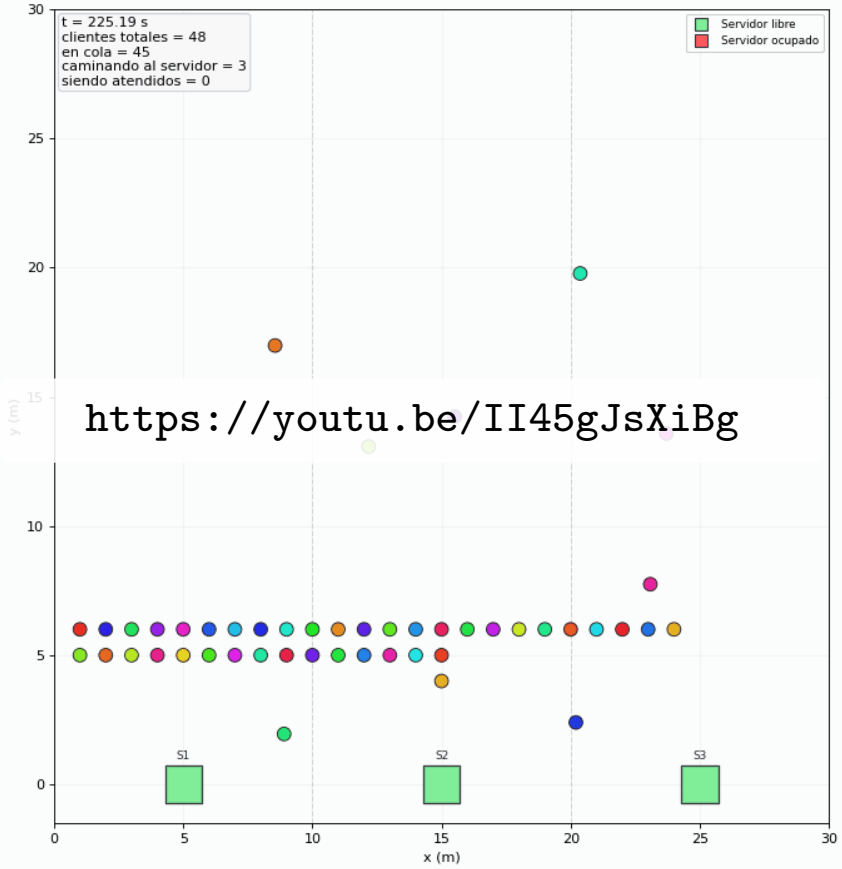
$t_1 = 1\,\mathrm{s} \mid t_2 = 3\,\mathrm{s} \mid k \in \{1,\ 4,\ 8\}$
(izq. a der.)

Variación de k — Modalidad A — Tiempo de permanencia



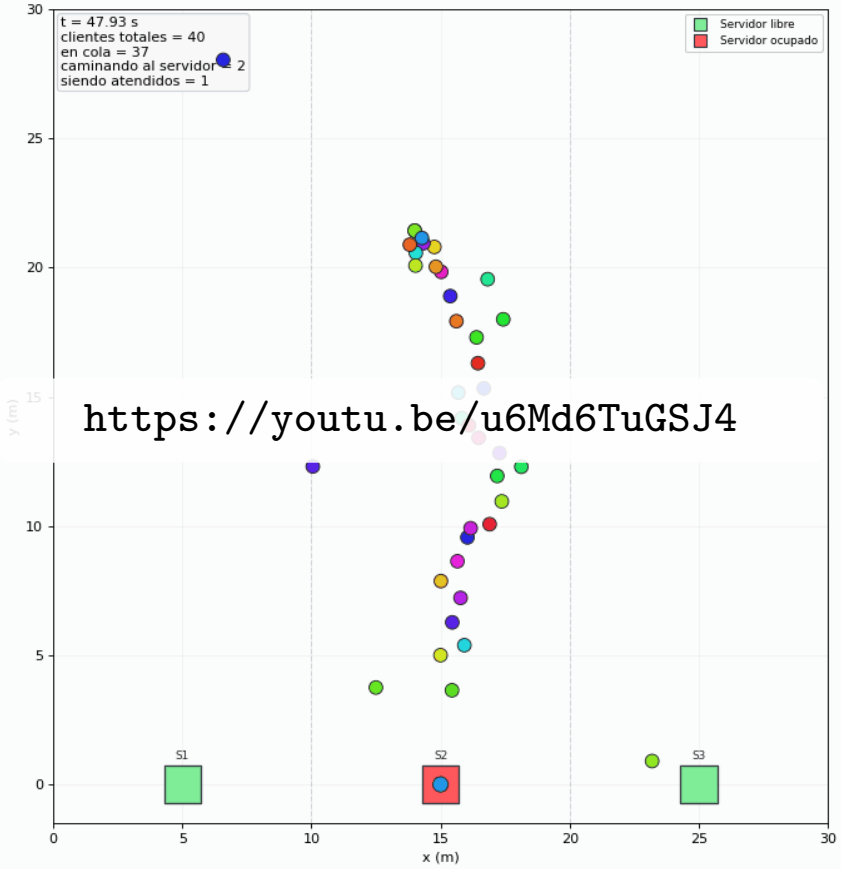
$t_1 = 1\,\mathrm{s}$ | $t_2 = 3\,\mathrm{s}$ | runs = 30

Animaciones — Modalidad B



https://youtu.be/II45gJsXiBg

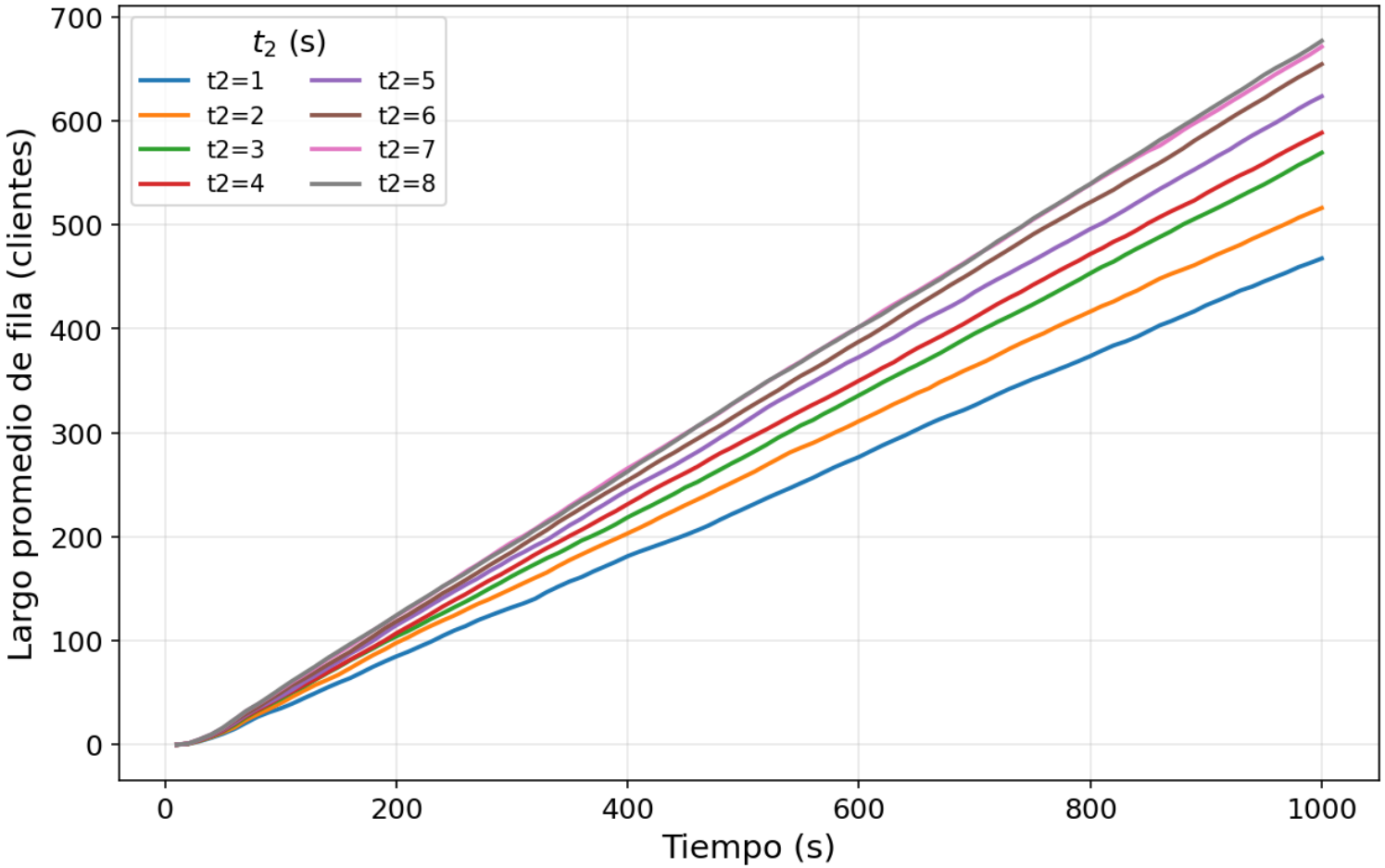
$t_1 = 2 \quad | \quad t_2 = 1 \quad | \quad k = 3$



https://youtu.be/u6Md6TuGSJ4

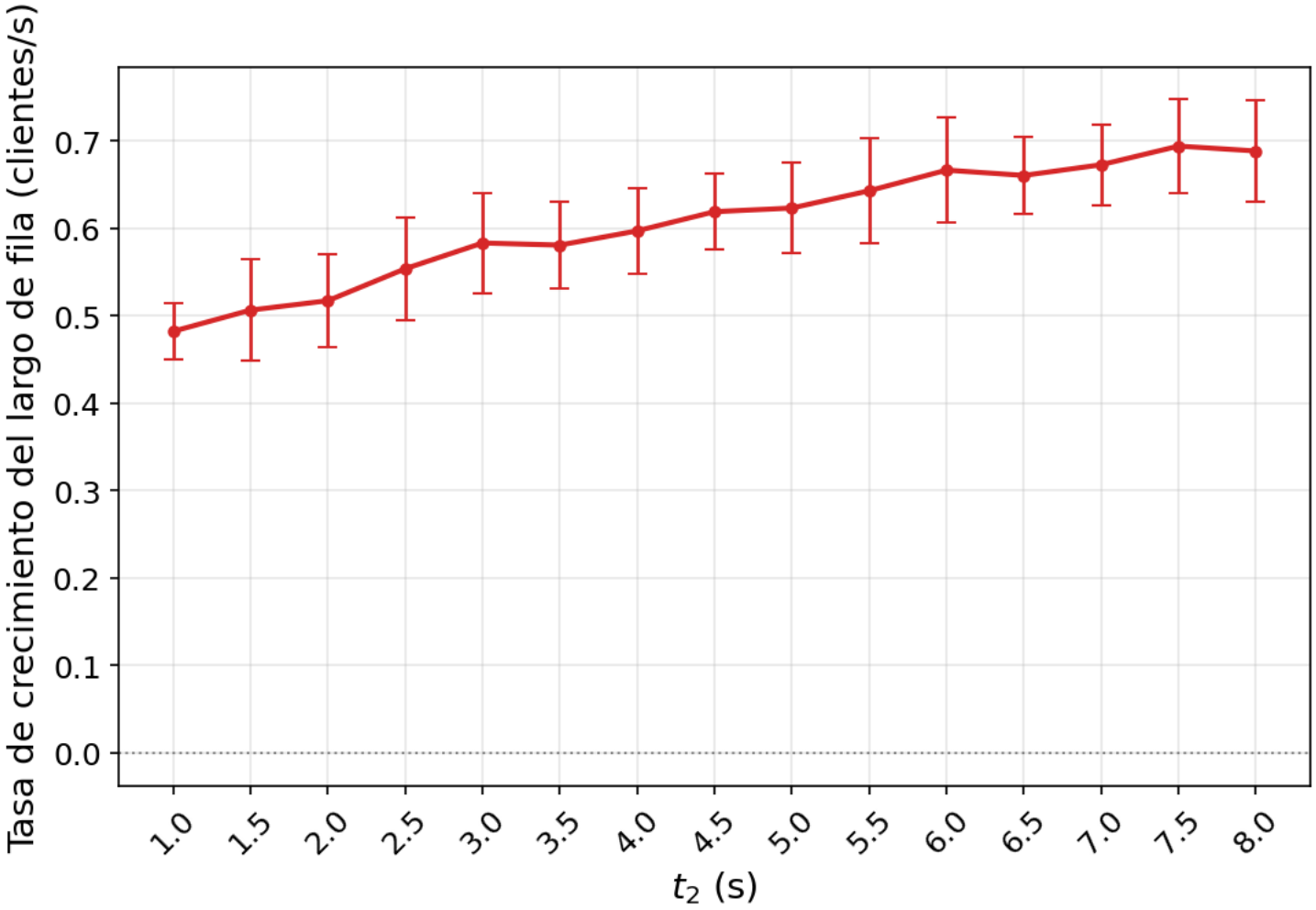
$t_1 = 1 \quad | \quad t_2 = 3.5 \quad | \quad k = 3$

Variación de t_2 — Modalidad B — $\langle L \rangle$ vs tiempo



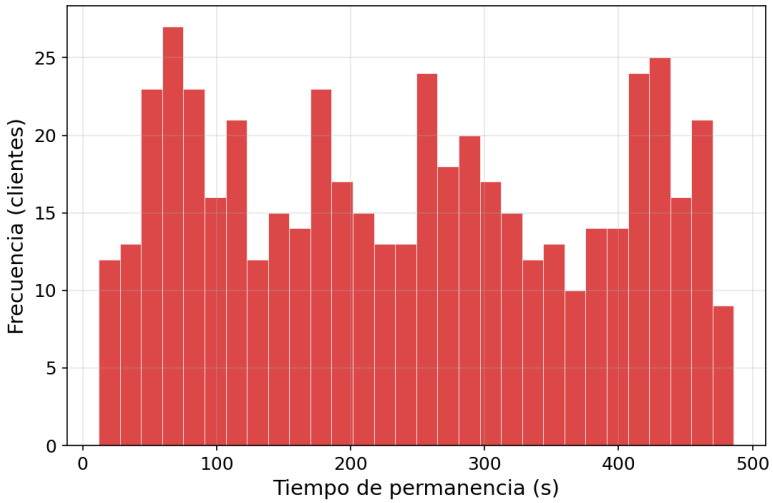
$t_1 = 1\,\mathrm{s}$ | $k = 5$ | runs = 30

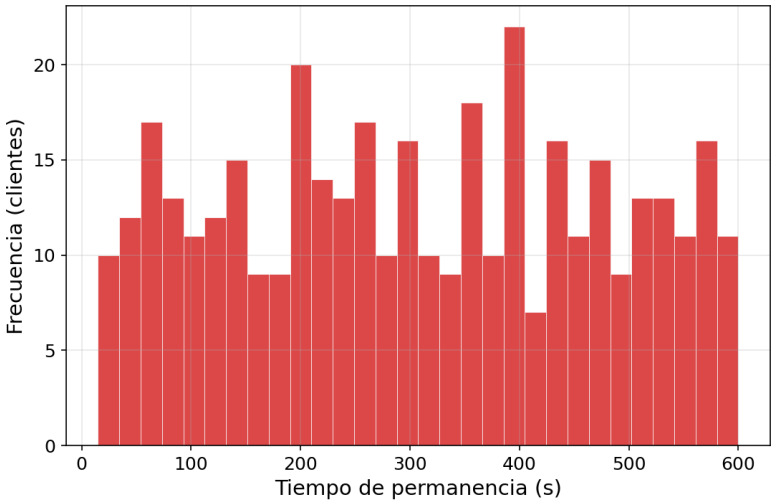
Variación de t_2 — Modalidad B — Tasa de crecimiento

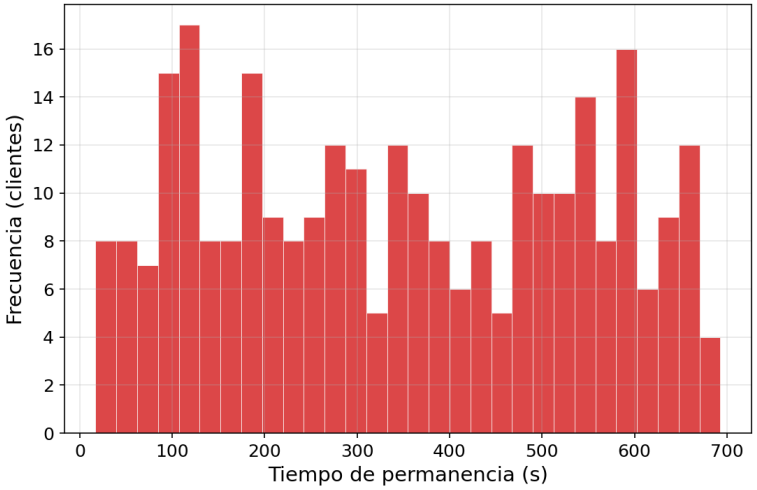


$t_1 = 1\,\mathrm{s}$ | $k = 5$ | runs = 30

Variación de t_2 — Modalidad B — Histogramas de permanencia

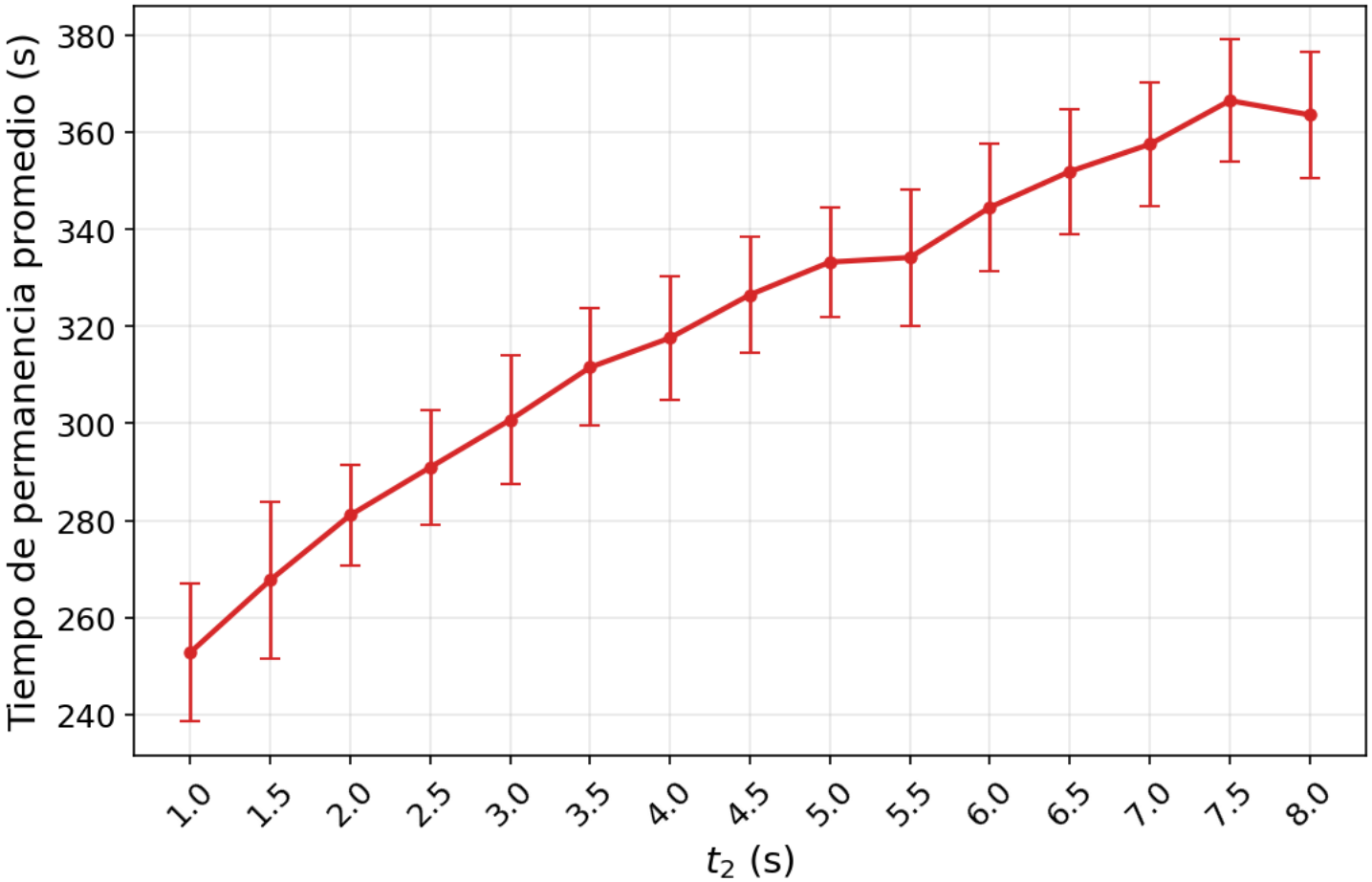






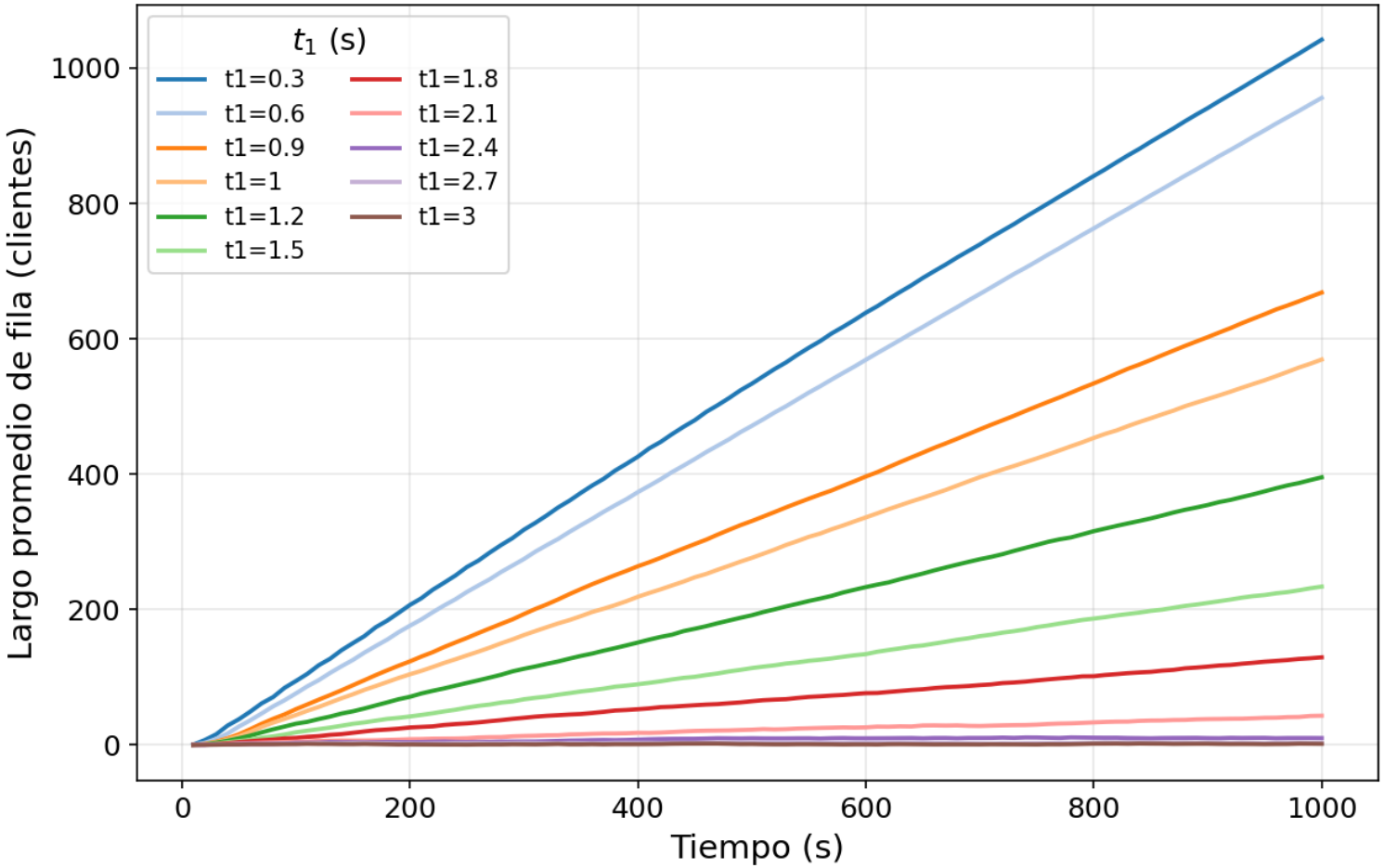
$t_1 = 1\,\mathrm{s} \mid k = 5 \mid t_2 \in \{1,\ 4,\ 8\}\,\mathrm{s}$
(izq. a der.)

Variación de t_2 — Modalidad B — Tiempo de permanencia



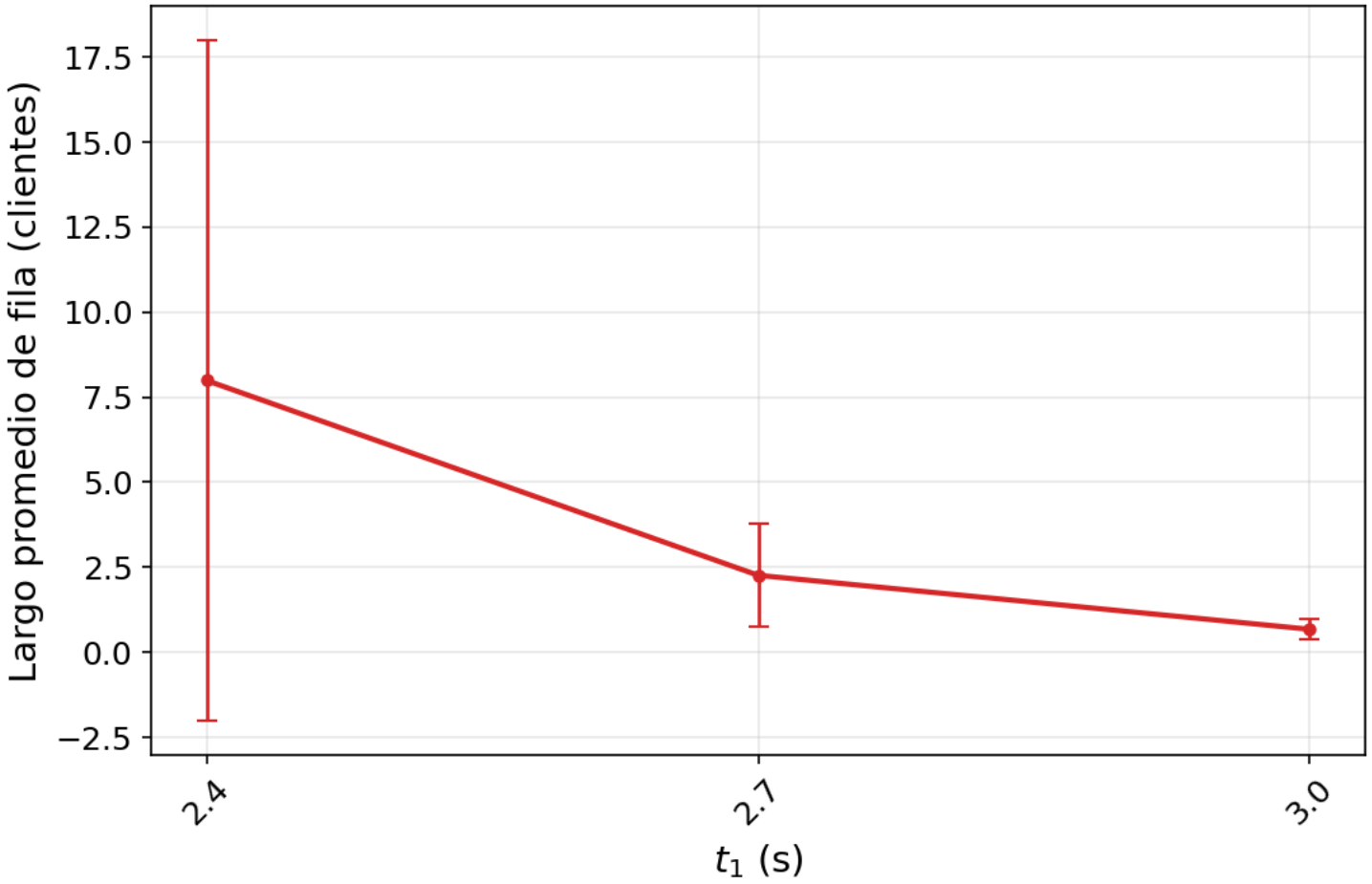
$t_1 = 1\,\mathrm{s}$ | $k = 5$ | runs = 30

Variación de t_1 — Modalidad B — $\langle L \rangle$ vs tiempo



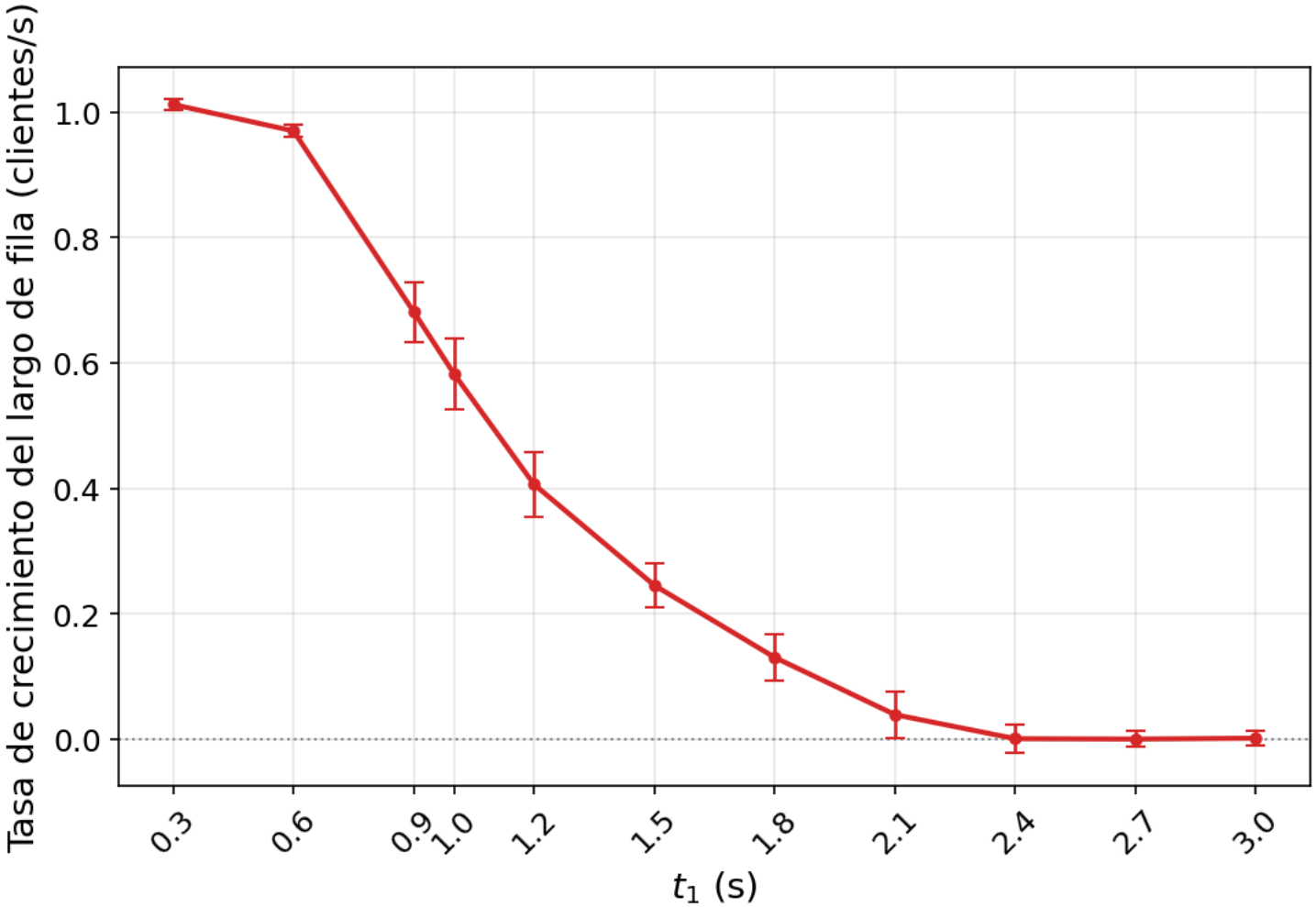
$t_2 = 3\,\mathrm{s}$ | $k = 5$ | runs = 30

Variación de t_1 — Modalidad B — Largo de la fila



$t_2 = 3\,\mathrm{s}$ | $k = 5$ | runs = 30

Variación de t_1 — Modalidad B — Tasa de crecimiento



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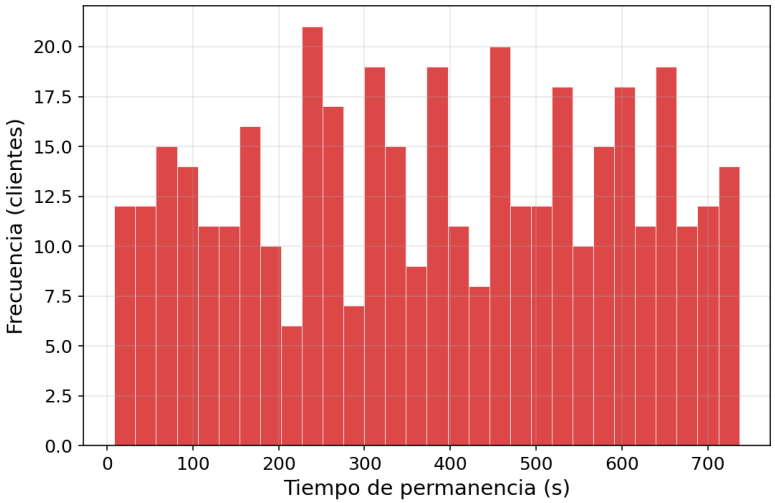
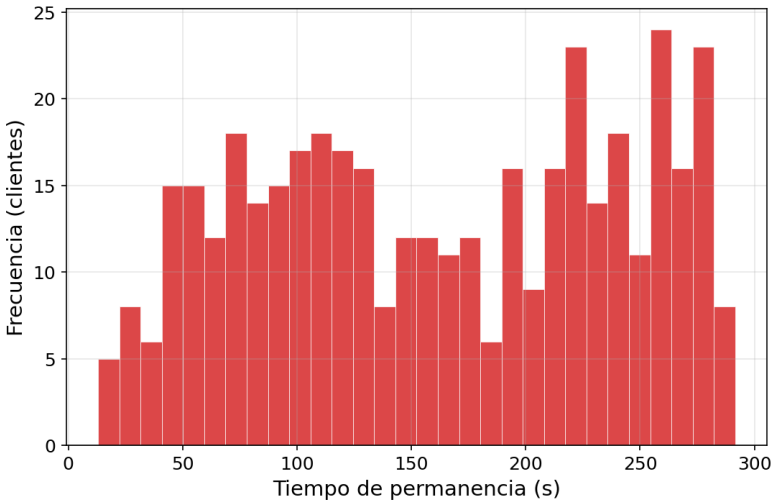
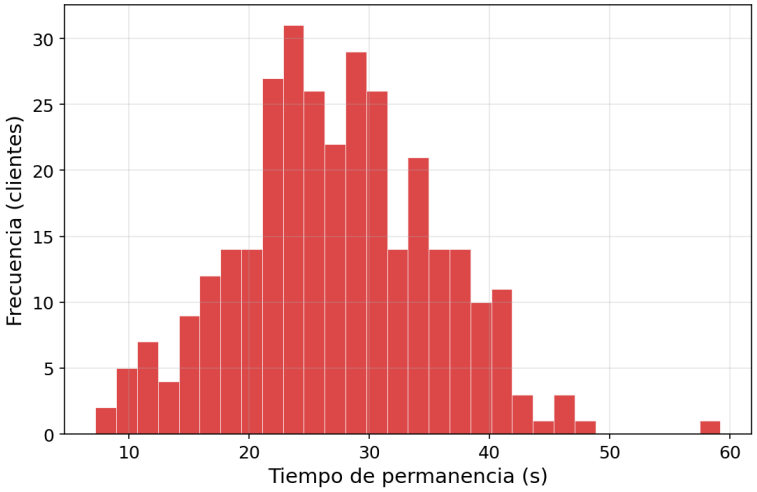
Variación de t_1 — Modalidad B — Histogramas de permanencia

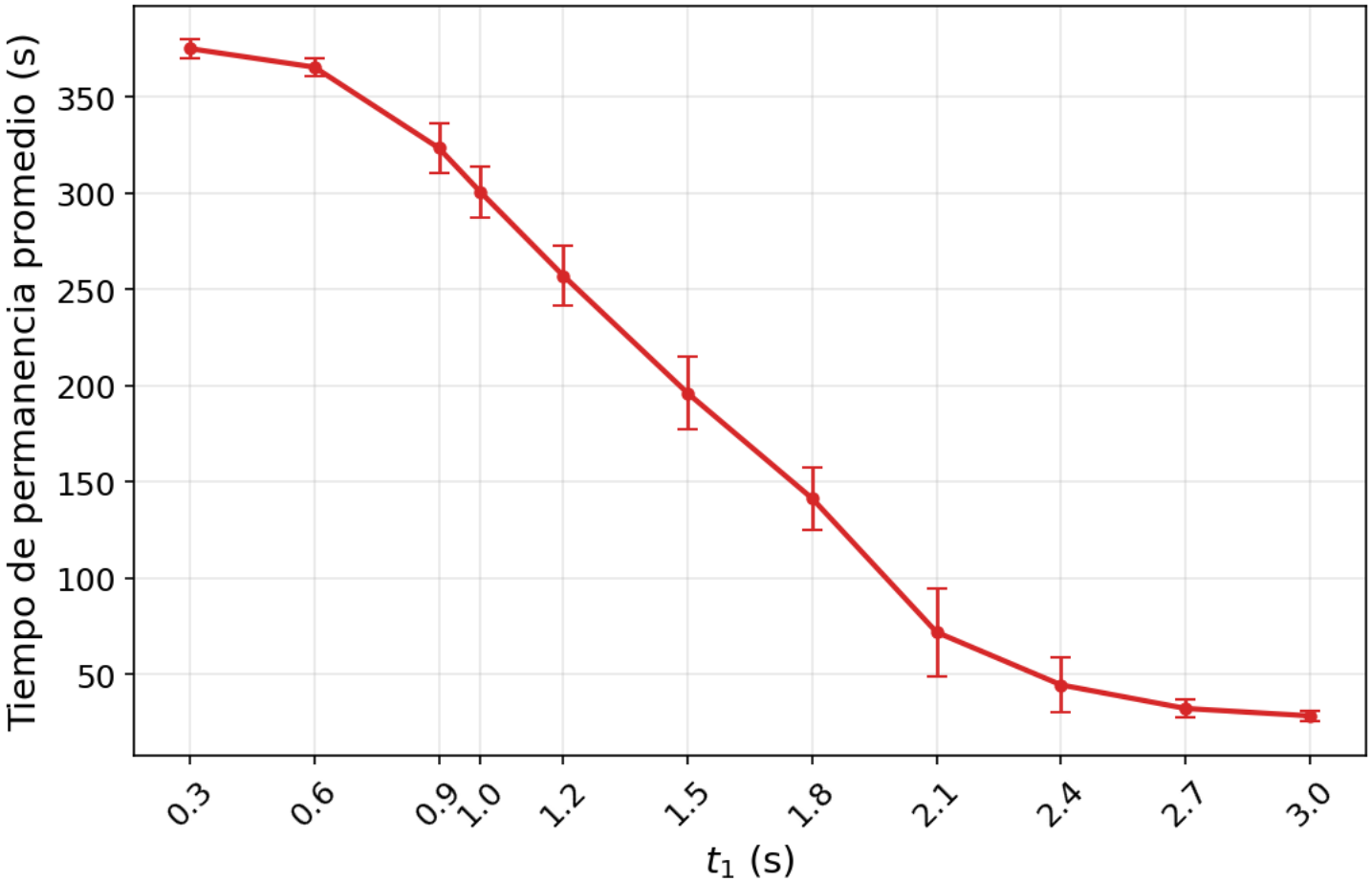






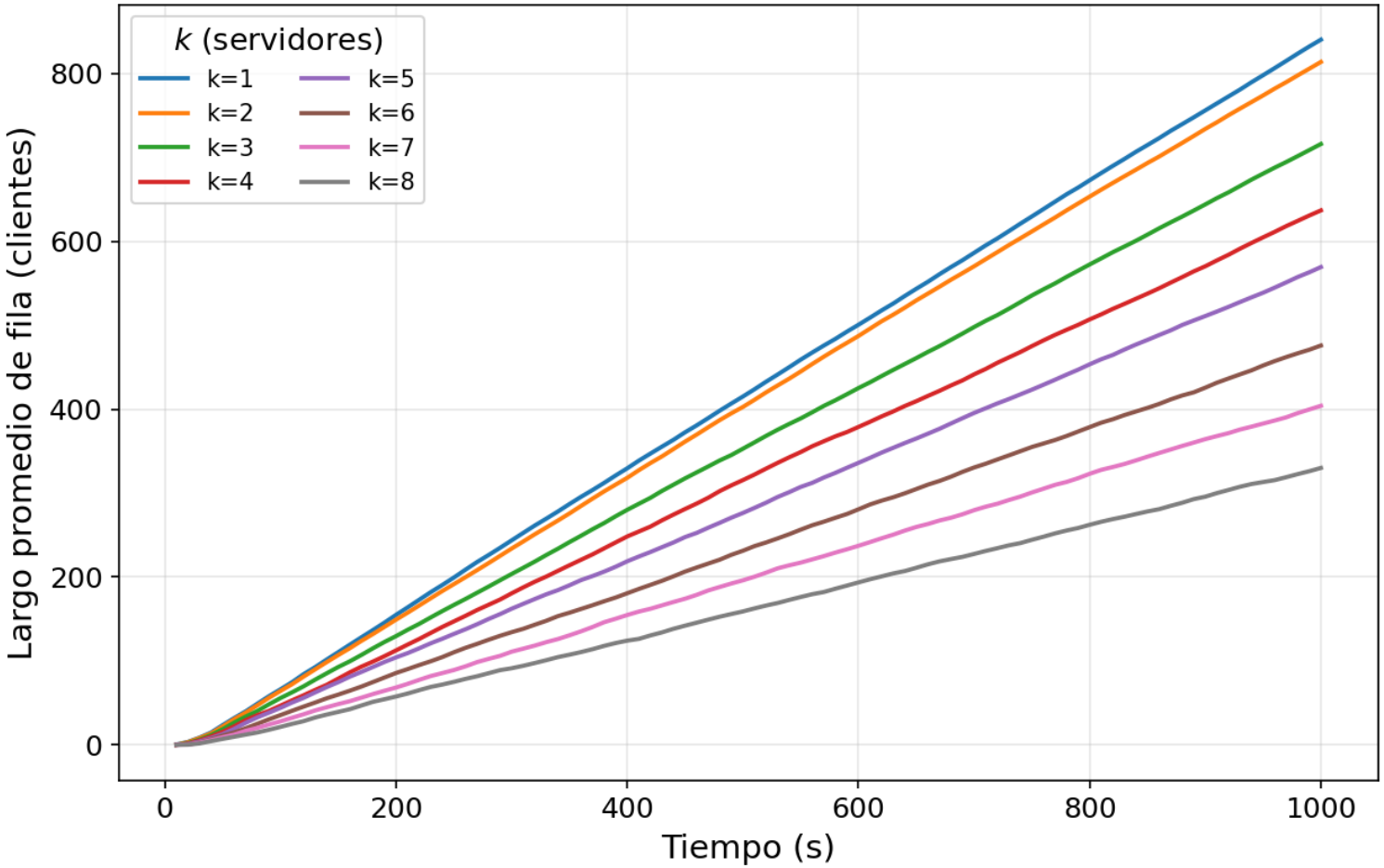
$t_2 = 3\,\mathrm{s}$ | $k = 5$ |
$t_1 \in \{0.3,\ 1.8,\ 3\}\,\mathrm{s}$ (izq. a der.)

Variación de t_1 — Modalidad B — Tiempo de permanencia



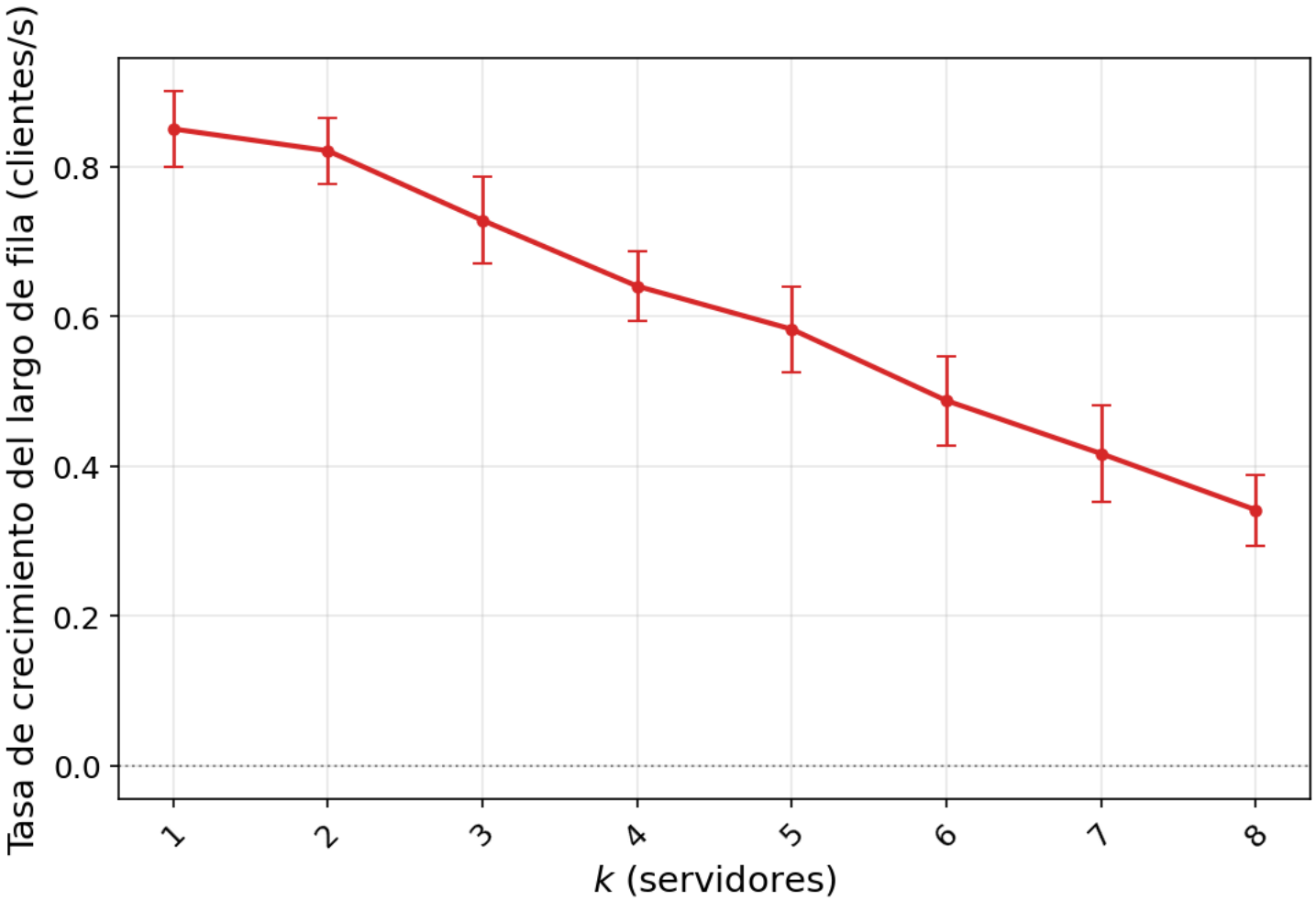
$t_2 = 3\,\mathrm{s}$ | $k = 5$ | runs = 30

Variación de k — Modalidad B — $\langle L \rangle$ vs tiempo



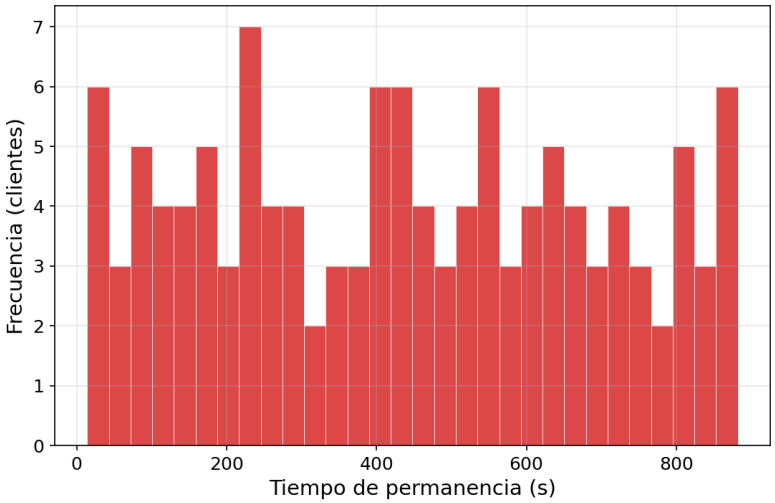
$t_1 = 1\,\mathrm{s}$ | $t_2 = 3\,\mathrm{s}$ | **runs = 30**

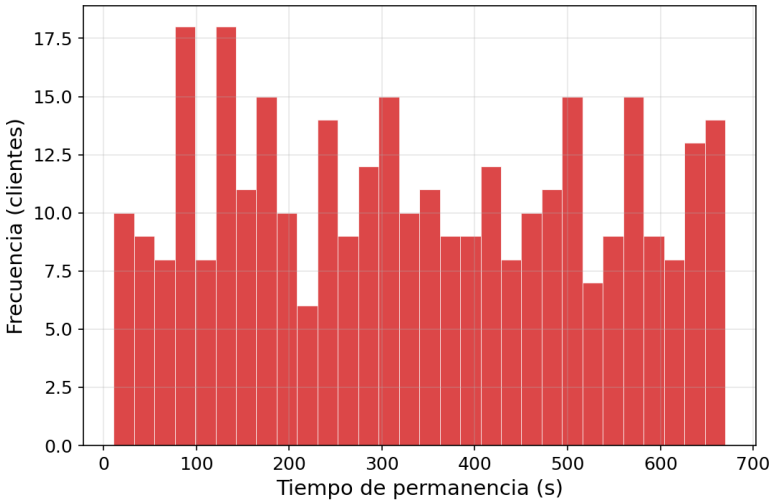
Variación de k — Modalidad B — Tasa de crecimiento

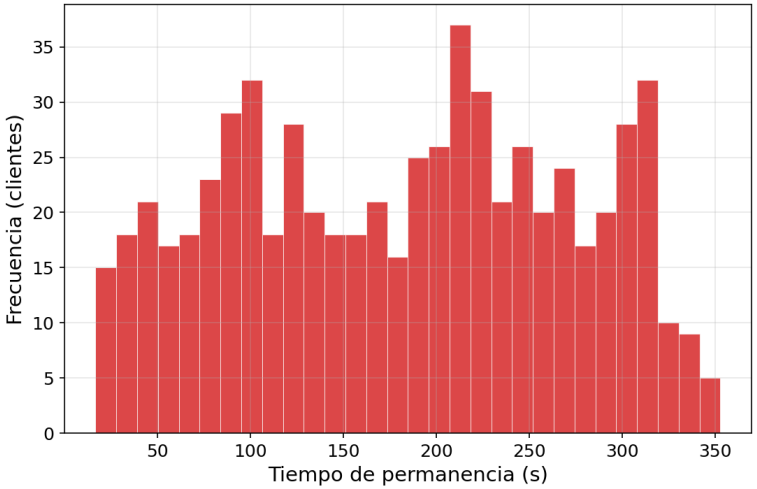


$t_1 = 1\,\mathrm{s}$ | $t_2 = 3\,\mathrm{s}$ | runs = 30

Variación de k — Modalidad B — Histogramas de permanencia

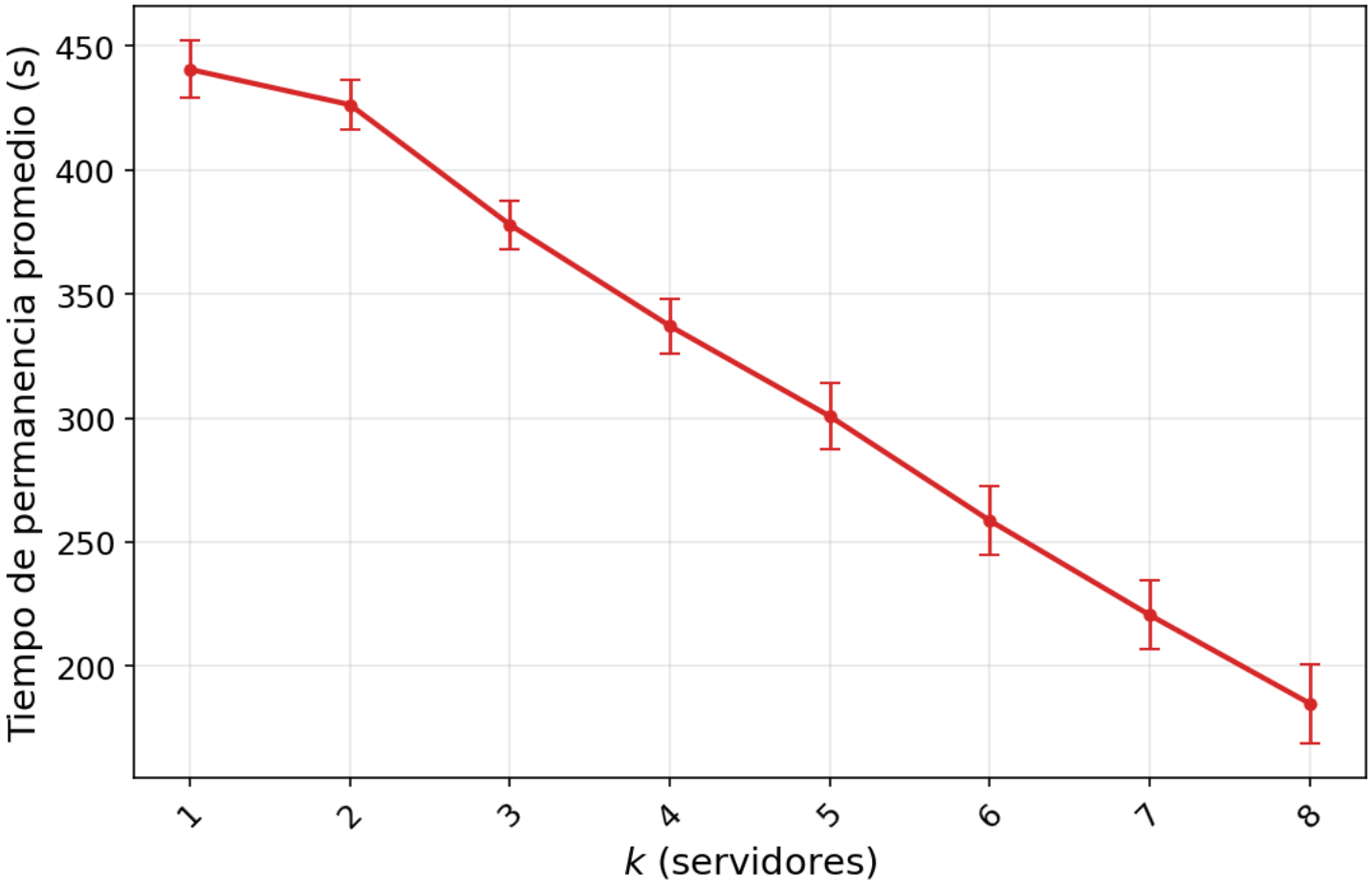






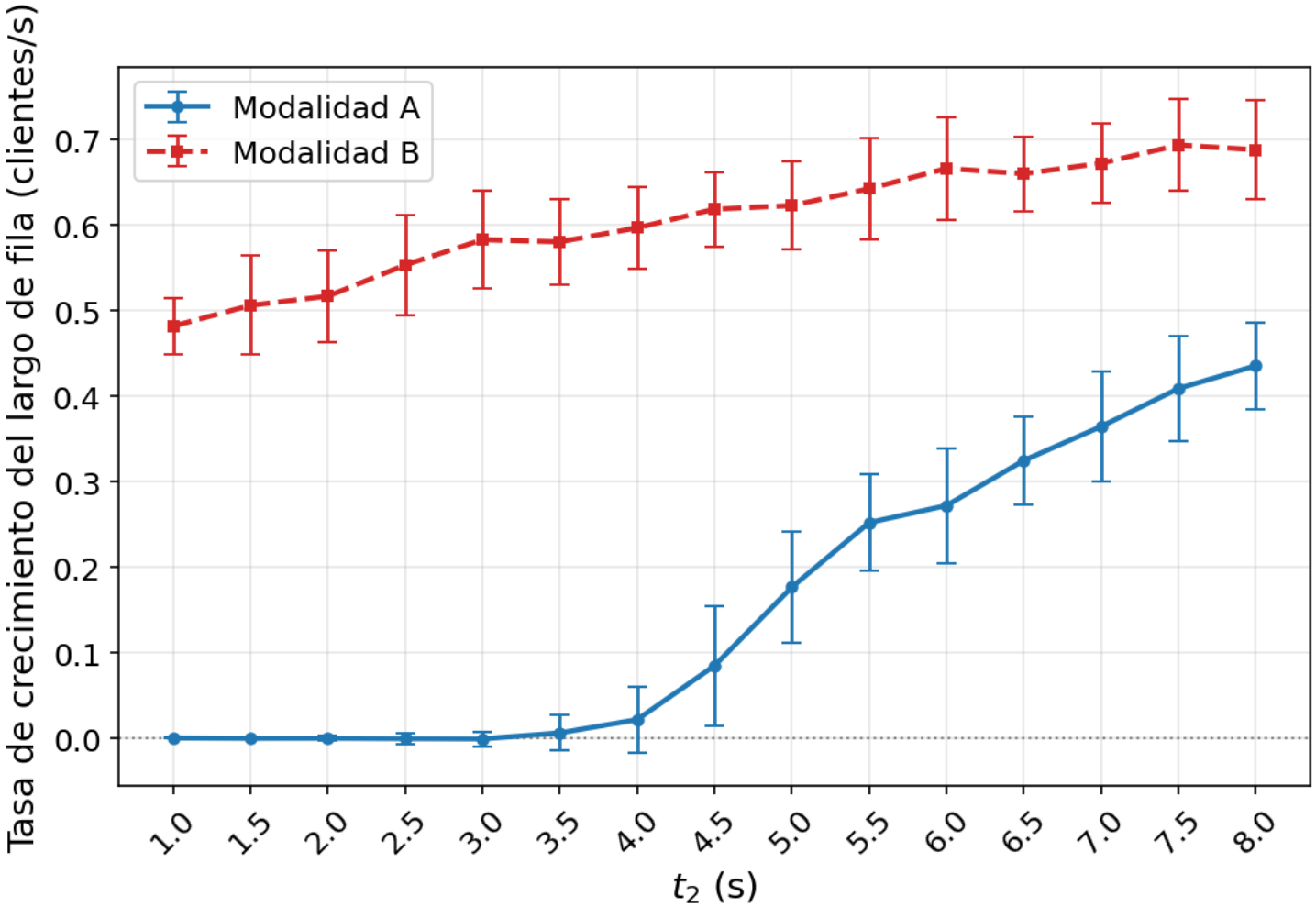
$t_1 = 1\,\mathrm{s} \mid t_2 = 3\,\mathrm{s} \mid k \in \{1,\ 4,\ 8\}$
(izq. a der.)

Variación de k — Modalidad B — Tiempo de permanencia



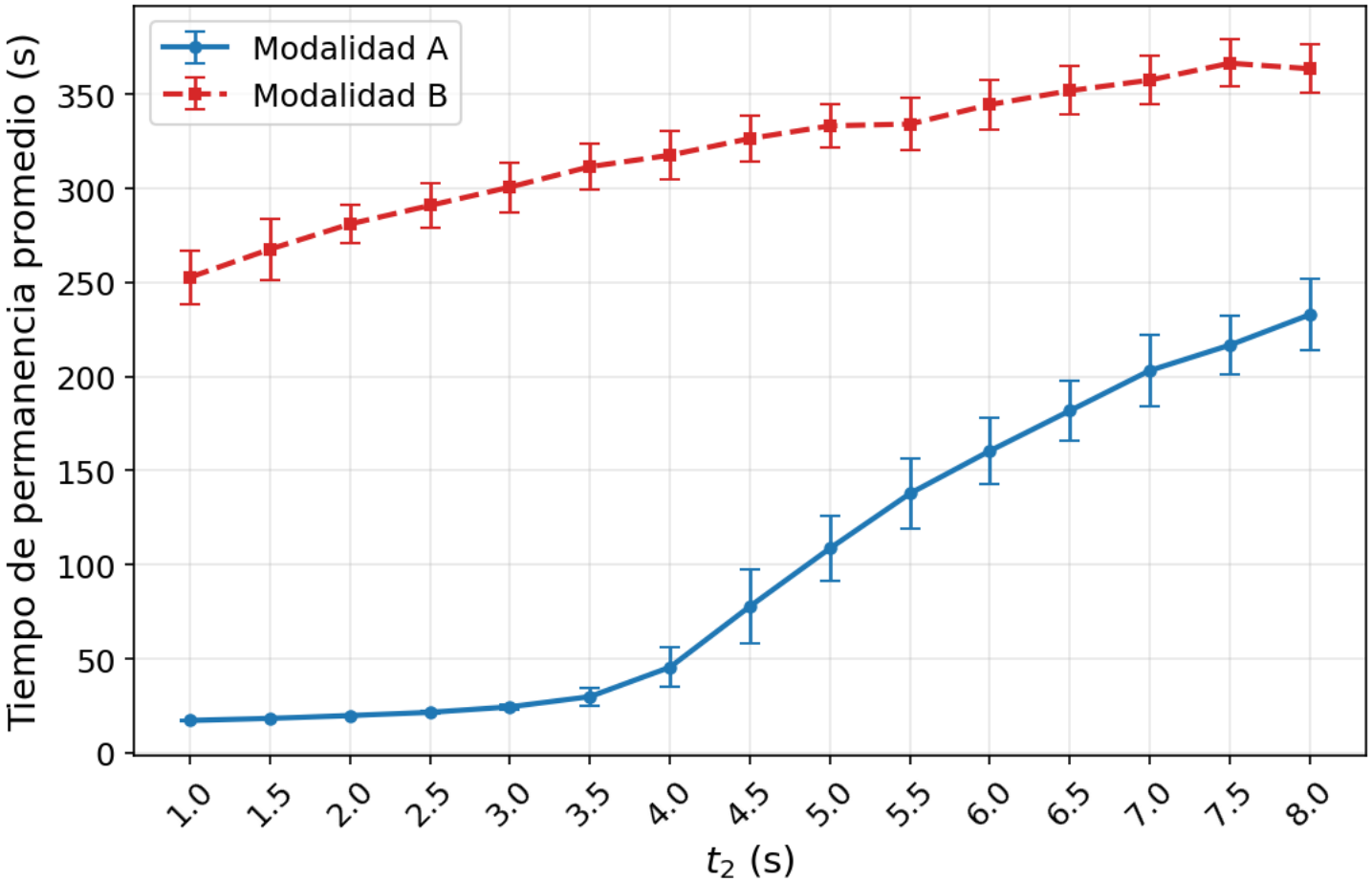
$t_1 = 1\,\mathrm{s}$ | $t_2 = 3\,\mathrm{s}$ | runs = 30

Comparación A vs B — t_2 — Tasa de crecimiento



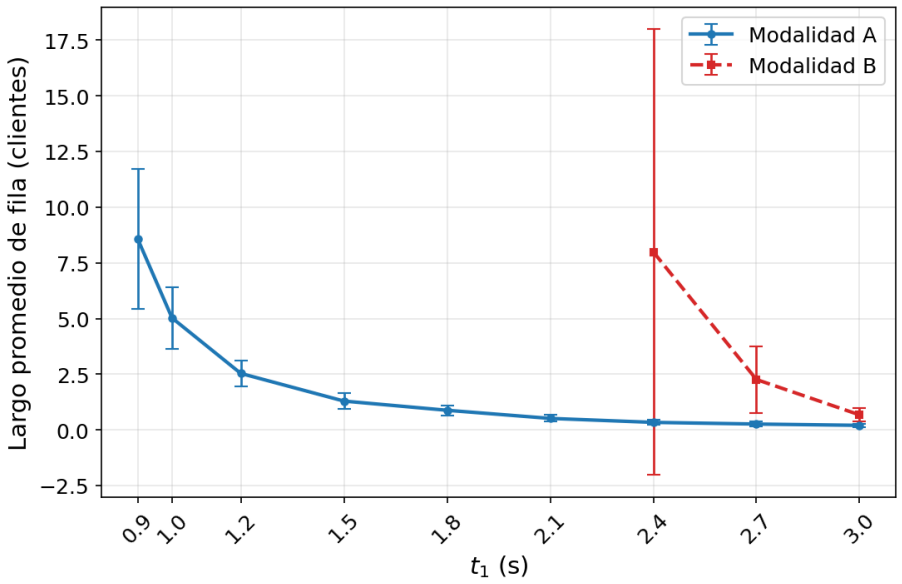
$t_1 = 1\,\mathrm{s}$ | $k = 5$ | runs = 30

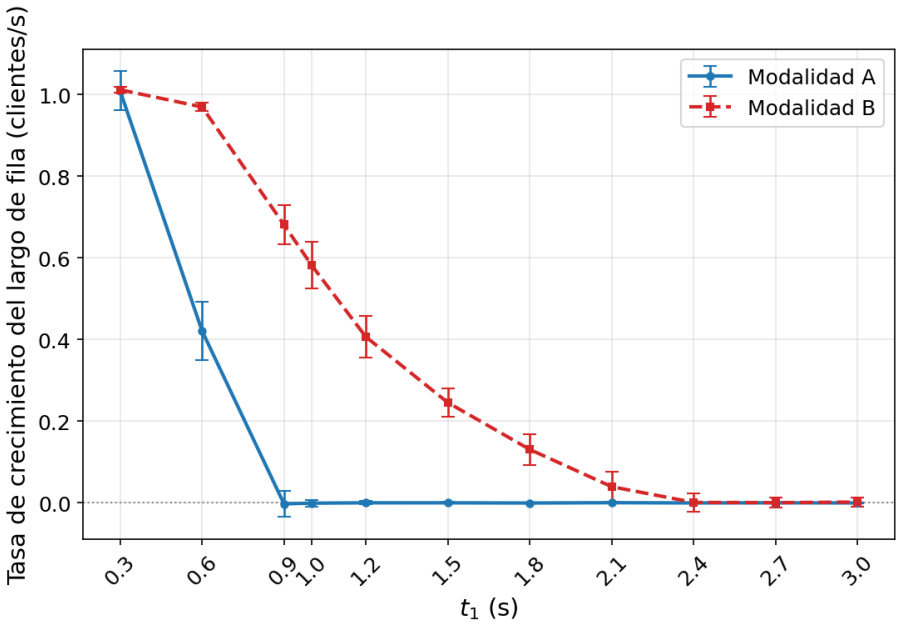
Comparación A vs B — t_2 — Tiempo de permanencia



$t_1 = 1\,\mathrm{s}$ | $k = 5$ | runs = 30

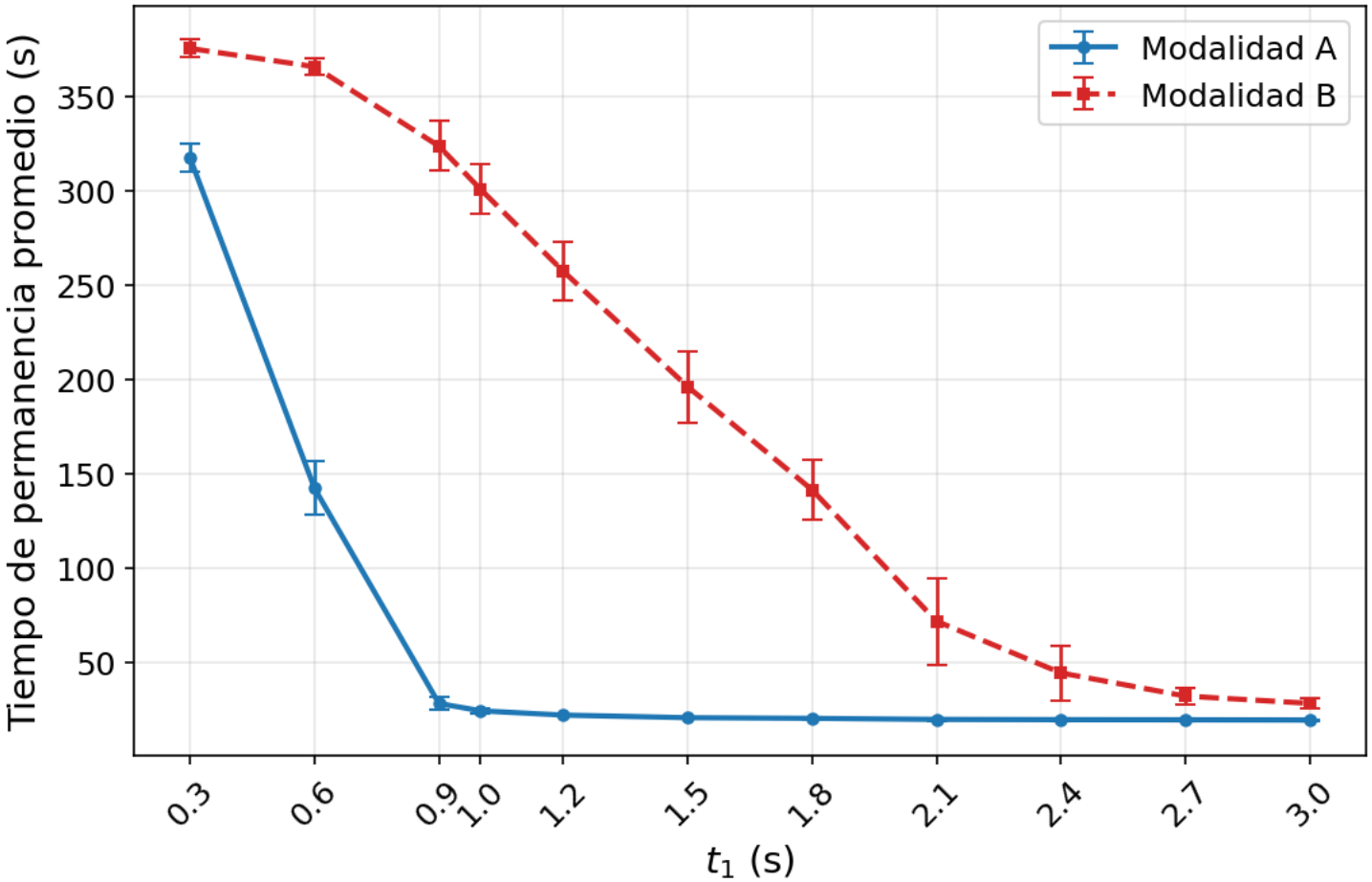
Comparación A vs B — t_1 — Largo de fila y tasa de crecimiento





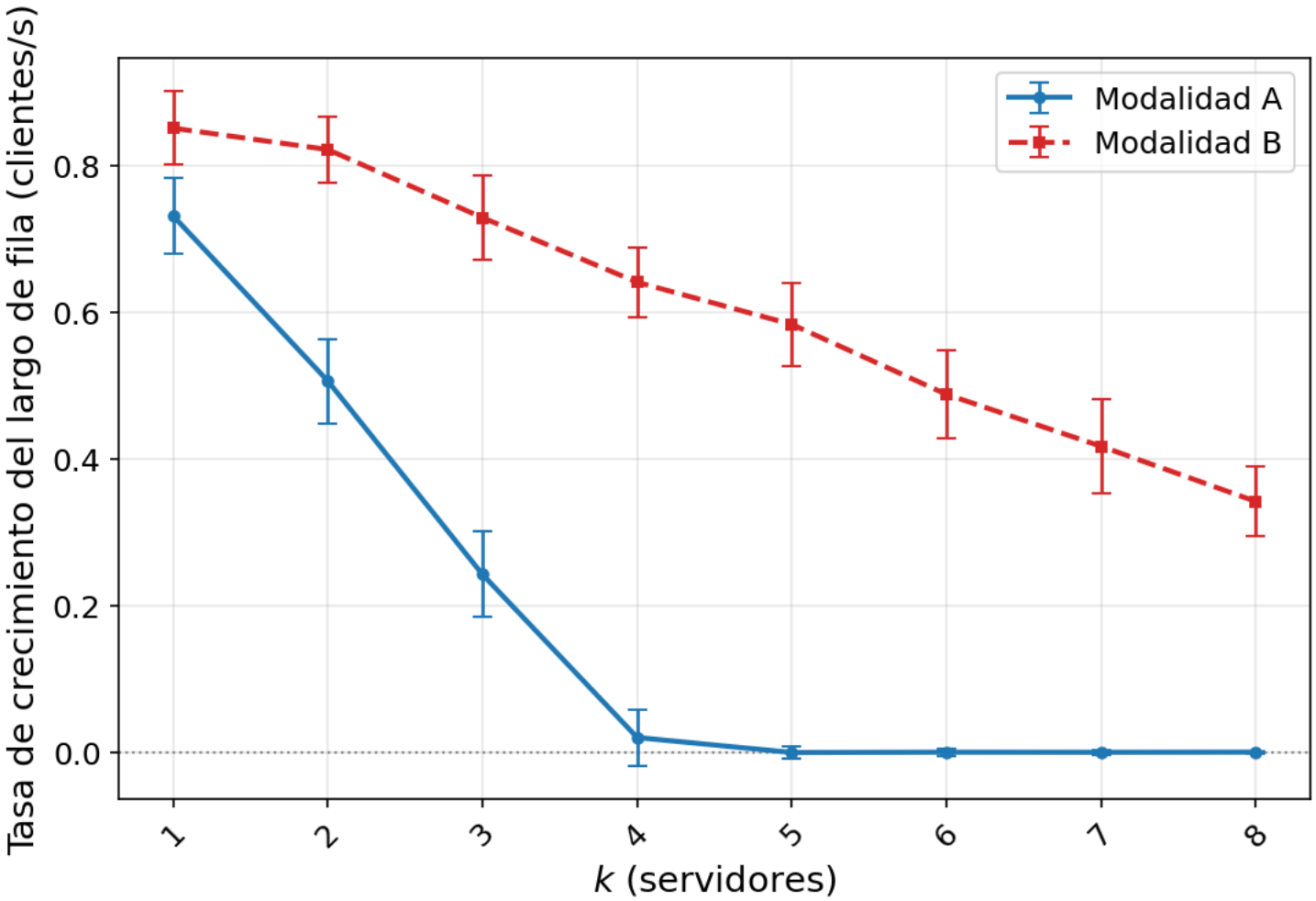
$t_2 = 3\,\mathrm{s}$ | $k = 5$ | runs = 30

Comparación A vs B — t_1 — Tiempo de permanencia



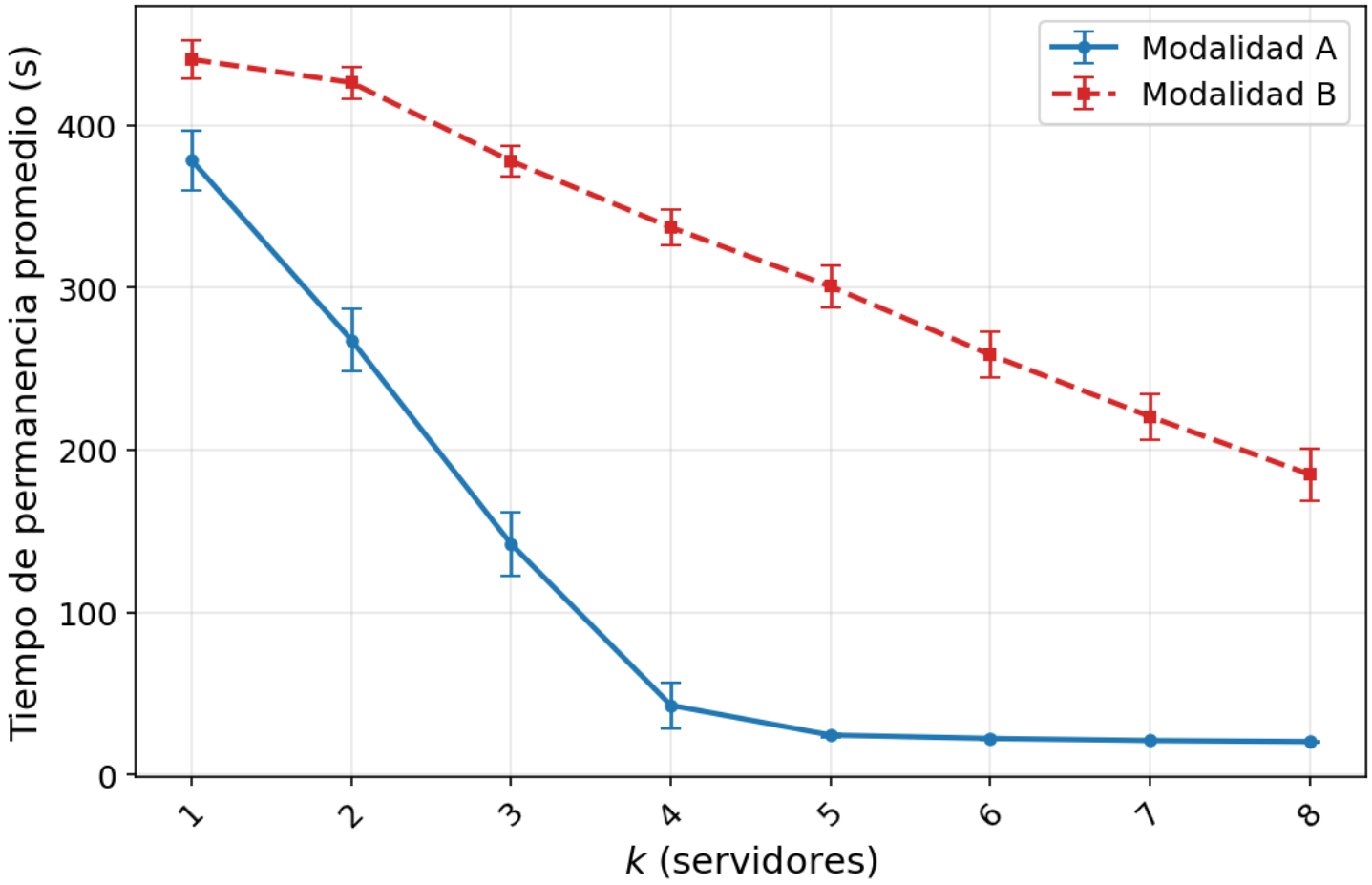
$t_2 = 3\,\mathrm{s}$ | $k = 5$ | runs = 30

Comparación A vs B — k — Tasa de crecimiento



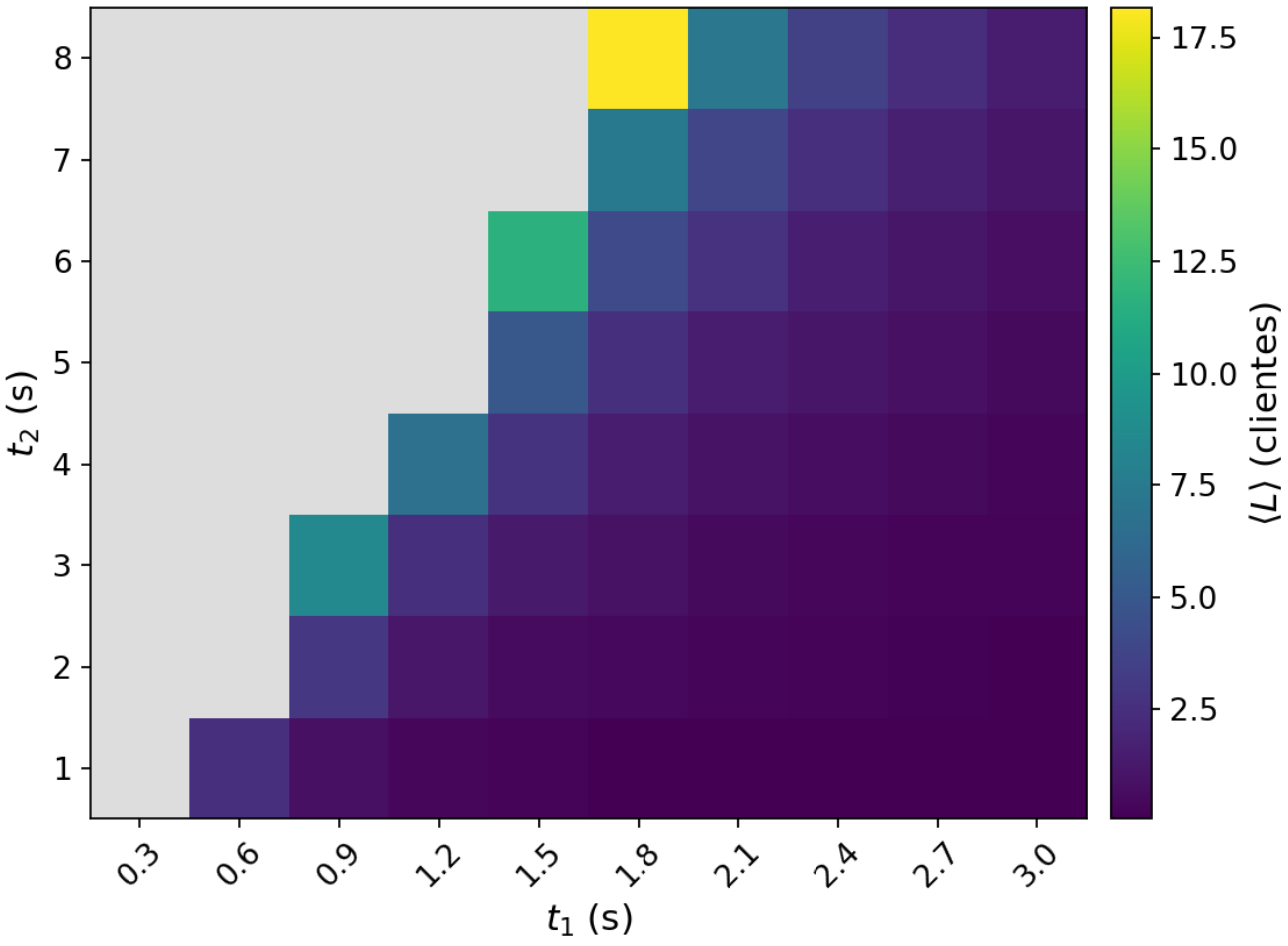
$t_1 = 1\,\mathrm{s}$ | $t_2 = 3\,\mathrm{s}$ | runs = 30

Comparación A vs B — k — Tiempo de permanencia



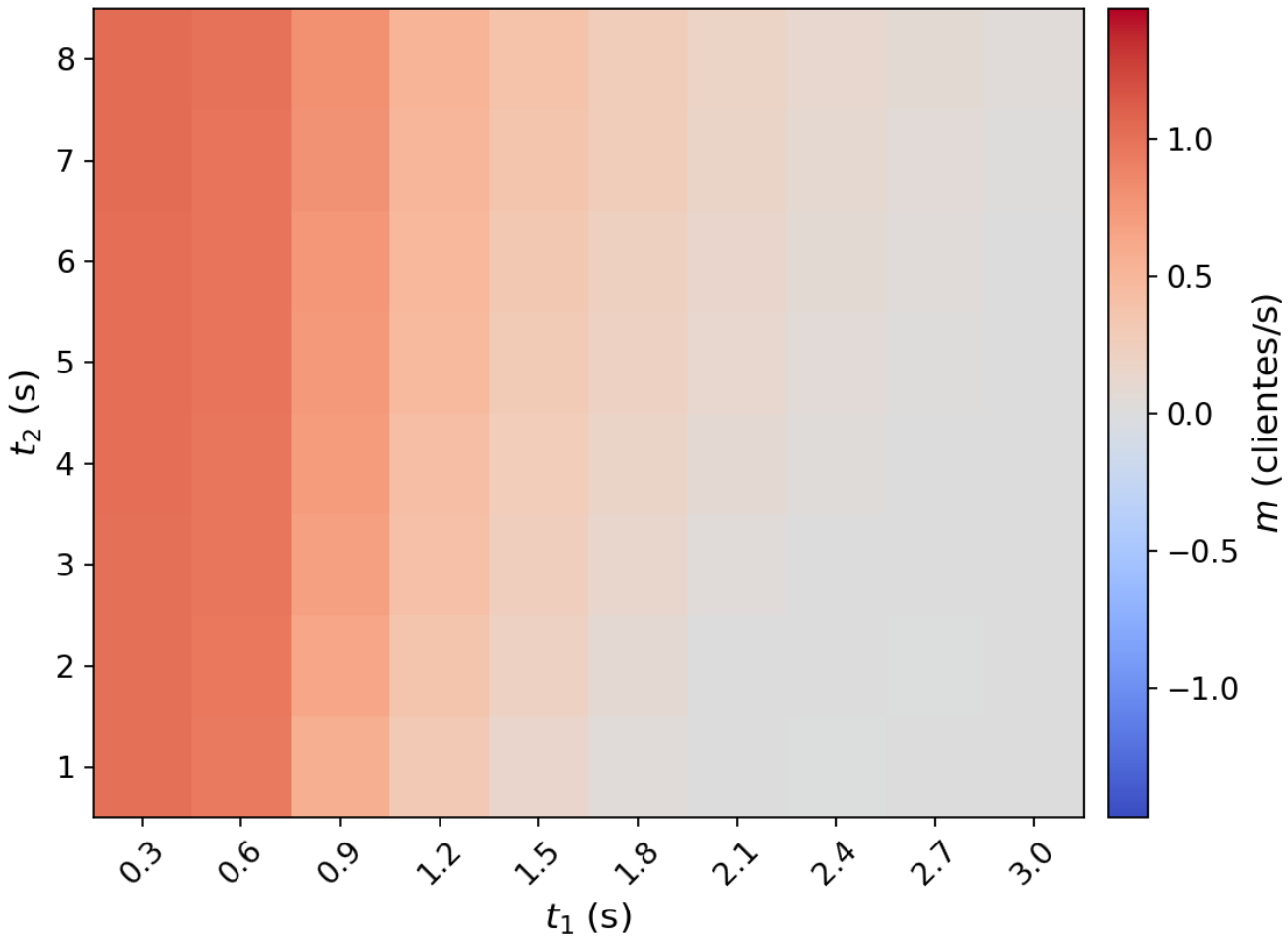
$t_1 = 1\,\mathrm{s}$ | $t_2 = 3\,\mathrm{s}$ | runs = 30

Mapa de Calor — Largo de la fila — Modalidad A



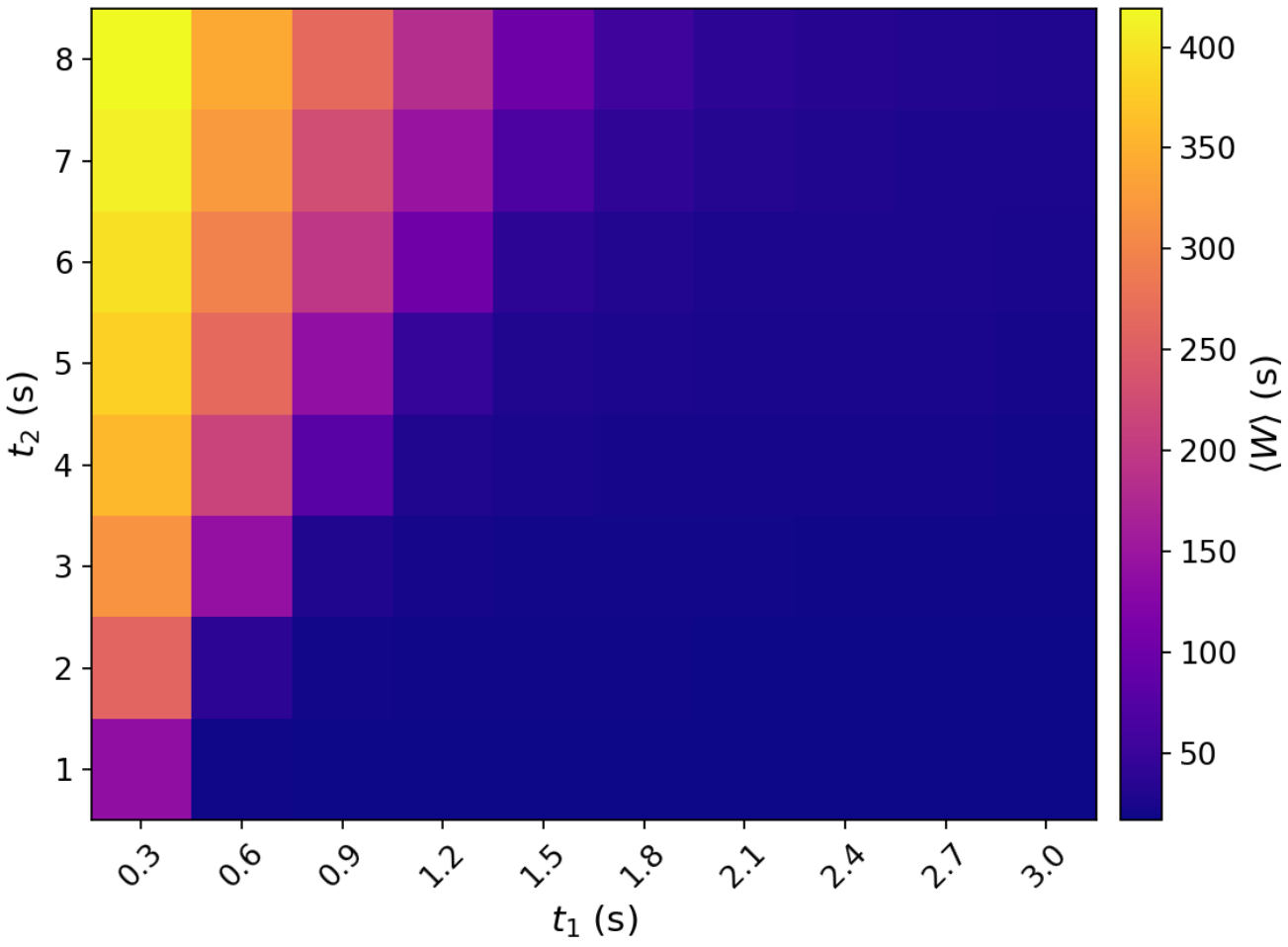
k=5 | runs = 30

Mapa de Calor — Tasa de crecimiento — Modalidad B



k=5 | runs = 30

Mapa de Calor — Tiempo de permanencia — Modalidad A



k=5 | runs = 30

Conclusiones

Conclusiones

- Para la fila guiada tipo serpentina, la modalidad A presenta menores largos de fila y menores tiempos de permanencia que la modalidad B en los tres barridos analizados.
- Al aumentar t_2, la congestión y la permanencia crecen en ambas modalidades; en B la tasa de crecimiento permanece positiva en todo el rango mostrado.
- Al aumentar t_1, ambas modalidades reducen la congestión, pero A alcanza valores cercanos a cero para largos de fila más rápidamente que B.
- Al aumentar k, ambas modalidades mejoran, aunque la reducción de cola y permanencia es mucho más marcada en A que en B.

Muchas gracias